

## Effective Health Aid: Evidence from Gavi's Vaccine Program<sup>†</sup>

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*Gavi, the Vaccine Alliance has provided over US\$16 billion in funding for vaccination in low- and middle-income countries since 1999. We exploit differential timing across countries and vaccines to estimate the effects of this public health initiative. We find that Gavi's support for a vaccine increased coverage rates by 2–5 percentage points across all vaccines and by 10–20 percentage points for newer vaccines. We also find that Gavi's support reduced child mortality from related causes by 1 child per 1,000 live births. We estimate these improvements saved around 1.5 million lives at a cost of about US\$9,000 per life saved. (JEL F35, F53, I12, I15, I18, J13, O15)*

One of the most pressing questions in economics is what can be done to improve outcomes for the world's poorest citizens. Recent work documents the importance of investing in the health of low-income children (Hendren and Sprung-Keyser 2020; Currie 2020, Baird et al. 2016; Bleakley 2010). In this paper, we study one of the world's most substantial investments in child health, Gavi, the Vaccine Alliance, a primary source of funding and support for vaccination in the world's lowest-income countries.

Since its founding in 1999, Gavi has distributed over US\$16 billion in vaccine aid to low- and middle-income (LMI) countries. Gavi primarily funds the purchase of vaccine doses but also funds grants for health-system strengthening in recipient countries. At its founding, Gavi funded vaccines that were already available in lower-income countries: for example, the diphtheria, pertussis, and tetanus (DPT) and hepatitis B (HepB) vaccines. However, more recently, Gavi was instrumental in introducing two new vaccines that target pneumococcal and rotavirus diseases to lower-income countries (Kremer, Levin, and Snyder 2020). Previous literature on the effectiveness of Gavi aid has found mixed evidence but is more relevant for the preexisting vaccines; these new vaccines target the most common causes of death

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for children in developing countries, suggesting a greater potential to impact welfare. Gavi is also currently the major source of funding for developing countries to acquire COVID-19 vaccines.

The manner in which Gavi rolled out funding with differential timing across countries and vaccines allows for a systematic, causal evaluation of this funding on vaccine coverage rates and child mortality. In order to identify the impact of Gavi support, we exploit variation across countries, cohorts, and vaccines. While eligibility for Gavi support was determined by a country's level of per capita gross national income (GNI) in 1998, there is considerable variation within a country in when certain vaccines were introduced with Gavi support, as well as variation within a vaccine in when it was introduced across different countries. As we discuss further below in Section IIB, this variation stems partly from external forces (e.g., vaccine supply constraints, changes in region-specific World Health Organization (WHO) recommendations) as well as from internal constraints (e.g., the availability of funds to meet Gavi's cofinancing requirement). Our identification strategy compares coverage of vaccines funded by Gavi to vaccines not funded by Gavi in a particular country for a particular cohort. We use a triple-difference specification in which *country*  $\times$  *vaccine* fixed effects account for possibly endogenous decisions to fund certain vaccines in certain countries, *country*  $\times$  *cohort* fixed effects flexibly allow for time-varying country characteristics, and *vaccine*  $\times$  *cohort* fixed effects account for global time-varying vaccine characteristics (such as global supply or demand shocks).<sup>1</sup>

To develop some intuition for our identification strategy, consider the case of two African countries, Niger and Senegal. Senegal received funding from Gavi for the measles-rubella (MR) vaccine beginning in 2013, while Niger never received funding for the MR vaccine and did not receive funding for the measles vaccine until 2019. Neither country received Gavi funding for the Bacillus Calmette-Guérin (BCG) vaccine. According to survey estimates, Senegal increased coverage for measles-containing vaccines by 5.6 percentage points for the 2015 birth cohort relative to the 2011 birth cohort. For the same cohorts, Senegal increased coverage of the BCG vaccine by 1.8 percentage points. Subtracting these two measures, we could estimate the effect of Gavi's program as increasing measles coverage by 3.8 percentage points. However, it is not obvious that measles and BCG coverage rates would trend in the same manner absent Gavi intervention. To estimate the expected increase in measles coverage relative to BCG absent Gavi intervention, we use Niger as a control country. Niger increased measles and BCG coverage by the same amount (7.4 percentage points) for the same birth cohorts. Assuming that, absent Gavi funding, Senegal would have seen a similar zero difference in the measles coverage rate relative to the BCG coverage rate, we conclude that 3.8 percentage points is the correct estimate for Gavi's effect. Heuristically, our identification

<sup>1</sup>Our results are robust to an instrumental variables strategy with Gavi eligibility (determined before 2000) interacted with global Gavi funding for a given vaccine in a given year (minus own country receipts) as the instrument. This strategy addresses possible bias stemming from time-varying policy shifts on the part of a recipient country to support a specific vaccine that might be correlated with Gavi support.

strategy averages many of these natural experiments across birth cohorts, vaccines, and countries to determine the effect of Gavi's program.

To provide further intuition for our estimation approach, we also briefly describe the method we use to correct for the bias that stems from heterogeneous effects and staggered treatment in standard difference-in-difference (DD) models (De Chaisemartin and d'Haultfœuille 2020; Goodman-Bacon 2021). We use the method developed by Borusyak, Jaravel, and Spiess (2024)—henceforth, BJS—which is straightforward to implement in our triple-difference framework. We aim to calculate a treatment effect for each treated observation: the outcome for the treated observation minus the counterfactual outcome (what would have happened absent Gavi funding). Using BJS, the counterfactual is estimated using our triple-difference specification only on untreated observations. The observation-level treatment effect is the observation outcome minus the fitted value from the counterfactual regression. Overall treatment effects are then calculated as simple averages of these treatment effects across all treated observations.

Using OLS and the BJS method, we estimate a 2–5 percentage point increase in coverage across all Gavi-funded vaccines, but the impacts are substantially larger, ranging from 10 to 20 percentage points, for the pneumococcal and rotavirus vaccines, two new vaccines that were mostly unavailable in these countries prior to Gavi's support. An event-study specification provides precise estimates of the increase in coverage with no evidence of preexisting differential trends, which supports our identifying assumptions.

Using a similar triple-difference strategy, we estimate the impact of the Gavi-funded introduction of a vaccine on mortality from related causes between the ages of 1 and 59 months, which we hereafter refer to as child mortality.<sup>2</sup> Mortality-by-cause data in LMI countries are often imputed due to the lack of high-quality vital registration data. To avoid these imputations, we use the subset of the Global Health Observatory (GHO) data (Perin et al. 2022a, b<sup>3</sup>) that is based on vital registry (VR) data. We compare trends in child mortality from causes related to Gavi-funded vaccines to trends in child mortality from other causes. We use a flexible specification with *country*  $\times$  *cause of death*, *country*  $\times$  *year*, and *cause of death*  $\times$  *year* fixed effects to allow for identification of these mortality effects. We find that child mortality from causes related to Gavi-funded vaccines falls by about 1 child per 1,000 live births for each vaccine introduced.<sup>4</sup> Notably, we find that these effects are driven by significant reductions in respiratory deaths and diarrheal deaths after Gavi funded the introduction of the pneumococcal and rotavirus vaccines.<sup>5</sup>

<sup>2</sup>We omit neonatal mortality since infants do not generally receive vaccinations at birth except for the HepB birth dose and BCG, which Gavi did not fund.

<sup>3</sup>Perin et al. (2022b) presents estimates from the research group whose work was the basis for the GHO estimates of the number of child deaths by cause (World Health Organization 2000–2017). As those GHO data end in 2017, we use the published estimates of Perin et al. (2022b) to extend the sample to 2019.

<sup>4</sup>Since the VR subset of the GHO covers a selected sample of countries, we also demonstrate that our results are robust to using the entire sample of countries in the GHO data as well as mortality-by-cause data from the Global Burden of Disease Study (Vos et al. 2020).

<sup>5</sup>We find no effects on deaths related to previously existing vaccines such as measles and meningitis, which could be due to coverage of those vaccines being closer to herd immunity, where further increases in coverage have smaller effects on mortality.

The pneumococcal vaccine protects against respiratory infections, while diarrhea resulting in severe dehydration is a common complication of rotavirus. As discussed in detail in Section IIIB, this mortality decline is in line with the 10–20 percentage point increase in coverage that we estimate for these vaccines.

Next, we show that coverage of non-Gavi funded vaccines continued to rise around the time that Gavi first began funding vaccines in a given country. This finding is important because, given our strategy of comparing coverage rates across vaccines, spillovers from Gavi-funded vaccines to nonfunded (control) vaccines have the potential to bias our results. In principle, these spillovers could be positive or negative and be demand- or supply-side. For example, a vaccination campaign for one vaccine may motivate households to inoculate their children against other diseases. Alternatively, countries could reallocate resources away from existing vaccines toward Gavi-supported vaccines in order to fund the cost sharing. Positive spillovers would suggest that our estimates are lower bounds on Gavi's true effect. Negative spillovers could result in our overestimating the effect of Gavi's programs, so we are reassured by the lack of evidence of these negative spillovers.

Using our estimates of Gavi's impact on child mortality, we conclude that aid through Gavi saved 1.6 million lives (90 percent confidence interval (CI): 561,000–2.75 million) at a cost of US\$8,750 (90 percent CI: \$5,035–\$23,353) per life saved between 2000 and 2019. However, we note that these estimates should be considered a lower bound on the full impact of Gavi on welfare. In this paper, we focus only on one benefit of vaccination: reduced child mortality. Some of the vaccines that Gavi funds target older ages (e.g., HPV) and have longer-term impacts on health. Gavi's support for strengthening health systems may have more far-reaching benefits as well.

Our paper is not the first to examine the effect of Gavi, but we contribute a new identification strategy and a number of novel and important results to this literature. Three evaluations estimate positive effects of Gavi funding (Lu et al. 2006; Hulls et al. 2010; Ikilezi et al. 2020) but use identification strategies susceptible to endogeneity concerns from time-varying differences across countries and vaccines. For example, if Gavi were to persistently target diseases whose mortality rates were falling for other reasons, their approaches would overestimate the effect of Gavi, while our *disease*  $\times$  *cohort* fixed effects will control for these unobservables. Moreover, Lu et al. (2006) and Hulls et al. (2010) use older data on fewer recipient countries and fewer funded vaccines. In contrast, Jaupart, Dipple, and Dercon (2019) find positive effects of Gavi using an event-study methodology. However, they consider only one event, the founding of Gavi in 1999–2000, and compare countries eligible for Gavi funding to countries not eligible for Gavi funding based on GNI per capita. Accordingly, they are unable to exploit the differential timing of Gavi funding across countries, time, and vaccines, which is key to our identification strategy. They estimate large reductions in infant and under-5 mortality (over 12 deaths per 1,000 live births), which are possibly driven by differential trends between Gavi-eligible and Gavi-ineligible countries. Concurrently, Dykstra et al. (2019) argue that most Gavi aid just replaced money already spent on vaccination using a regression discontinuity design that exploits the GNI per capita eligibility threshold. While well identified, their study examines only the impact on coverage

rates (not mortality). Moreover, their results are applicable only to countries near the eligibility cutoff (the least poor of the Gavi-eligible countries) and are not applicable to newer vaccines like pneumococcal or rotavirus (for which they argue that the eligibility cutoff was not strictly enforced). Given that lower respiratory infections and diarrheal diseases are a significant cause of mortality among children in poor countries, examining the impact of these vaccines is essential to understanding the full impact of Gavi. We find that Gavi's funding of these two vaccines had substantial impacts on child mortality.

Our paper also informs the literature on the effectiveness of foreign aid. Seminal early work on this question by Burnside and Dollar (2000) documented a positive link between foreign aid and economic growth when recipient countries have good underlying economic policies. However, subsequent work argued that these and similar results were sensitive to various assumptions (Easterly, Levine, and Roodman 2004; Roodman 2007). These disputes remain, even among studies that rigorously consider the causal impact of aid. Rajan and Subramanian (2008) and Werker, Ahmed, and Cohen (2009) fail to find beneficial effects. Other recent papers find a positive, if sometimes modest, overall effect on economic growth (Clemens et al. 2011; Galiani et al. 2017). When turning from economic growth to health, the literature often finds positive effects, but uncertainty remains. For example, multiple papers find that aid reduces infant mortality (Arndt, Jones, and Tarp 2015; Mishra and Newhouse 2009; Kotsadam et al. 2018). However, Mukherjee and Kizhakethalackal (2013) and Williamson (2008) find no—or, at best, limited—impacts on infant and child mortality, life expectancy, and immunization. There is also evidence that aid increases conflict (Nunn and Qian 2014; Crost, Felder, and Johnston 2014), limiting the potential positive impact of aid on health.

One way forward in resolving this uncertainty may lay in rigorous evaluation of specific foreign aid programs with a tight link between aid and targeted outcomes (Bourguignon and Sundberg 2007; Easterly 2003; Banerjee et al. 2007; Banerjee and Duflo 2011; Qian 2015). Our study of the impact of Gavi follows this path. Indeed, successful vaccination campaigns should be tightly linked with outcomes such as vaccine coverage rates and child mortality from vaccine-preventable diseases. Moreover, Gavi is a significant source of foreign aid, having distributed more than US\$16 billion in funding for vaccines and vaccine support systems since it was founded in 1999.

The paper proceeds as follows. Section I provides background on Gavi, while Section II presents our empirical strategy and the data we use on Gavi funding, vaccine coverage, and cause-specific child mortality rates. Section III presents our results, Section IV offers a discussion of cost-effectiveness, and Section V concludes.

## I. Background on Gavi

Gavi, the Vaccine Alliance, was created in 1999 by its core partners the Bill and Melinda Gates Foundation, the WHO, UNICEF, and the World Bank. While the Gates Foundation provided initial funding for the alliance, sovereign donors promptly provided additional support. The United Kingdom was the first country to provide funding to the alliance in 2000, followed quickly by Denmark, Norway, Sweden, the



Netherlands, and the United States, and later joined by Canada, Ireland, France, and Luxembourg. By 2005 the alliance had US\$1.67 billion in donations, about half of which came from sovereign countries (Chee et al. 2008). Between 2000 and 2017, Gavi distributed about 40 percent of all foreign aid for immunization (Ikilezi et al. 2021). The aim of the partnership was to increase vaccine take-up, particularly in developing countries. The alliance hoped to accomplish this goal despite the inability of a predecessor organization, the Children's Vaccine Initiative, to make significant progress on increasing global immunization because of insurmountable disagreements among its various partners.

At the first meeting of the alliance, the board decided to limit Gavi support to only the lowest-income countries, defined as those with World Bank measured per capita incomes at or below US\$1,000 (GNI as of 1998). At the time, this rule made 74 countries eligible to apply for Gavi support, but 4 countries (Albania, Bosnia and Herzegovina, China, and Turkmenistan) soon exceeded the threshold and stopped receiving funding (Kallenberg et al. 2016). Since then, Gavi has revised upward the eligibility threshold, which, as of 2020, stands at US\$1,630.<sup>6</sup>

An important goal of Gavi is advancing sustainable vaccination financing. The alliance aims for countries to continue their vaccination programs even after they are no longer receiving funding from Gavi. To accomplish this goal, Gavi uses a tiered cofinancing policy described in Kallenberg et al. (2016). Initially, recipient countries contribute US\$0.20 per dose, and Gavi funding covers the remaining cost. The countries continue to pay this cost until their per capita income exceeds the World Bank low-income-country threshold (US\$1,085 as of 2021). Once a country's income surpasses this level, their cofinancing requirement increases 15 percent per year until their per capita income surpasses the current Gavi eligibility threshold. At that point, their cofinancing requirement increases linearly to 100 percent over a period of 5 years. Even after a country no longer receives Gavi funding postgraduation, it is eligible to purchase vaccines through UNICEF at Gavi prices, which in some cases are substantially lower than open market prices.

As explained in Dykstra et al. (2019), it is useful to categorize Gavi-funded vaccines into two broad categories. The first group of vaccines are long-standing, inexpensive vaccines that were commonly found in developing countries prior to the founding of Gavi. These vaccines include the DPT and the HepB vaccines. Prices for these vaccines range from US\$0.10 to US\$0.60.<sup>7</sup> Gavi has also modernized the vaccine programs of recipient countries. For example, it has replaced the DPT vaccine with the pentavalent vaccine, which protects against DPT, HepB, and *Haemophilus influenzae* type b (Hib). The pentavalent vaccine prices have ranged from US\$3.6 in 2004 to a low of US\$1.2 in 2020.

The second group of vaccines consists of the rotavirus and pneumococcal vaccines, two new vaccines which were largely unavailable or present in only a small

<sup>6</sup>Three countries (Cuba, the Democratic People's Republic of Korea, and Ukraine) were deemed eligible in 1999 despite possibly having income above this threshold because Gavi questioned the accuracy of their available GNI data (Gavi 1999). Our data list 77 countries that have ever received Gavi funding since its founding, including South Sudan, Timor-Leste, and Kiribati, which were later added. See <https://www.Gavi.org/types-support/sustainability/eligibility>.

<sup>7</sup>UNICEF vaccine procurement pricing data are available at <https://www.unicef.org/supply/pricing-data>.

number of lower-income countries prior to 2008 when Gavi began funding them. When launched, the rotavirus vaccine cost US\$5.00 per dose, and the pneumococcal vaccine cost US\$7 per dose. Gavi procured and purchased the pneumococcal vaccine via an innovative advanced market commitment described in Kremer, Levin, and Snyder (2020).

Overall, Gavi has funded ten different vaccines and spent over US\$16 billion on country programs since its founding.<sup>8</sup> Most of the money Gavi has distributed since 2000 (\$13 billion out of the \$16 billion) has gone to the actual purchasing of vaccine doses. However, Gavi does provide support for vaccine delivery systems. Vaccine introduction includes one-off costs for training, planning, social mobilization, and a communication campaign, for example. In addition, while most of Gavi funding supports routine immunization for infants, Gavi also provides funding for catch-up campaigns that target older children (measles and rubella, for example) and even adults (meningococcal meningitis and yellow fever, for example).

## II. Empirical Strategy

### A. Empirical Specifications

Our empirical strategy relies on variation in Gavi support across countries, time, and vaccines. We estimate a triple difference with a complete set of fixed effects. Specifically, we include *country*  $\times$  *vaccine*, *country*  $\times$  *cohort*, and *vaccine*  $\times$  *cohort* fixed effects (subsuming *country*, *vaccine*, and *cohort* fixed effects) in all our regressions.<sup>9</sup> We estimate

$$(1) \quad Y_{ivdt} = \beta_1 Post_{ivt} + \theta_{iv} + \theta_{it} + \theta_{vt} + \varepsilon_{ivdt},$$

where  $Y_{ivdt}$  is coverage of dose  $d$  of vaccine  $v$  in country  $i$  for the cohort born in year  $t$ , and  $Post_{ivt}$  is a dummy variable indicating that the cohort was born after Gavi provided funding for the introduction of vaccine  $v$  in country  $i$ . We cluster our standard errors by country. Dose refers to the course in the vaccine regime. For example, DTP is a three-dose vaccine regimen with a child receiving an injection at three distinct times in his or her childhood. We study the impact of Gavi support on all doses since Gavi funds all doses and every dose increases the vaccine's efficacy, but we provide a robustness check restricting the sample to the final dose of each vaccine.

The two-way interacted fixed effects eliminate bias from many possible factors. For example, the *country*  $\times$  *vaccine* fixed effects reduce bias from potentially endogenous decisions to fund certain vaccines in certain countries. The *country*  $\times$  *cohort* fixed effects ensure that we are relying only on variation across vaccines, comparing those that were or were not funded by Gavi in a particular country in a particular year. These fixed effects absorb all time-varying differences across countries that

<sup>8</sup>See <https://www.Gavi.org/sites/default/files/publications/progress-reports/Gavi-Progress-Report-2018.pdf> and <https://www.gavi.org/sites/default/files/2022-06/GAVI-Alliance-2021-Annual-Financial-Report.pdf>.

<sup>9</sup>Our data provide vaccine coverage rates by birth cohort, not by year. We use cohort to indicate birth cohort and to distinguish these fixed effects from year fixed effects in our analysis of child mortality.

might drive changes in vaccine coverage. Finally, the *vaccine*  $\times$  *cohort* fixed effects address bias from global time-varying differences across vaccines, such as global supply changes. Note that these two-way fixed effects eliminate bias if unobserved shocks have the same percentage point effect on all observations in a two-way cell because our coverage outcome variables are measured in percentages. Focusing on more similar countries and more similar vaccine types (new versus preexisting) makes our identification assumption more believable.

Due to the inclusion of these fixed effects, the primary remaining source of bias would be time-varying country-specific differences in coverage for specific vaccines. If coverage rates for vaccines that Gavi will fund in the future are trending differentially than other vaccines in a particular country relative to differences in trends between these same vaccines in non-Gavi-funded countries, this would signal a problem for our strategy. We examine possible preexisting trends with an event-study specification. This specification has the added benefit of allowing the impact of the Gavi-funded introduction of a vaccine to vary flexibly over time. We estimate

$$(2) \quad Y_{ivdt} = \sum_{l=-7}^{-2} \beta_l D_{ivt}^l + \sum_{l=0}^7 \beta_l D_{ivt}^l + \theta_{iv} + \theta_{it} + \theta_{vt} + \varepsilon_{ivdt},$$

where  $D_{ivt}^l$  indicates that the cohort born in year  $t$  in country  $i$  was born  $l$  years after (or, if  $l$  is negative, before) vaccine  $v$  was introduced. Estimates of  $\beta_l$  for  $l \geq 0$  tell us the impact of Gavi on coverage for cohorts born the year the vaccine was introduced and later, while estimates of  $\beta_l$  for  $l < 0$  tell us whether coverage for this vaccine relative to other vaccines was trending differently in the adopting countries relative to the nonadopting countries before Gavi funding was introduced. We estimate impacts using a 15-year window centered around the vaccine introduction, grouping 7 or more years out together on either side. The omitted time period is one year prior to introduction.

Our empirical specifications when studying child mortality from related, vaccine-preventable causes of death are similar, but complicated by the fact that multiple vaccines can affect some causes of death, limiting the within-country variation in relevant Gavi funding. Therefore, we link vaccines to causes of death in two ways. First, we exploit as much within-country variation as possible, by linking each Gavi-funded vaccine to the primary cause of death it targets. We estimate

$$(3) \quad Y_{ict} = \beta_1 Post_{ict} + \theta_{ic} + \theta_{it} + \theta_{ct} + \varepsilon_{ict},$$

$$(4) \quad Y_{ict} = \sum_{l=-5}^{-2} \beta_l D_{ict}^l + \sum_{l=-0}^5 \beta_l D_{ict}^l + \theta_{ic} + \theta_{it} + \theta_{ct} + \varepsilon_{ict},$$

where  $Y_{ict}$  is the number of deaths among children under the age of 5 per 1,000 live births (excluding deaths among infants less than one month old) in country  $i$  in year  $t$  from cause of death  $c$ ,  $Post_{ict}$  is a dummy variable indicating that Gavi has provided funding to country  $i$  prior to year  $t$  for a vaccine primarily affecting deaths from



cause  $c$ , and  $D_{ict}^l$  indicates that Gavi introduced funding for such a vaccine exactly  $l$  years before  $t$  (or, if  $l$  is negative, after  $t$ ). Due to data availability, we use a shorter time window of 11 years, grouping 5 or more years out together on either side. We include the full set of interacted fixed effects to ensure that our results are not driven by country-specific differences over time, country-specific differences across causes of death, or global time-specific differences across causes of death.

In addition, we estimate a specification in which each vaccine impacts every medically related cause of death by estimating

$$(5) \quad Y_{ict} = \beta_1 \text{NumVaccines}_{ict} + \theta_{ic} + \theta_{it} + \theta_{ct} + \varepsilon_{ict},$$

where  $\text{NumVaccines}_{ict}$  is the number of vaccines that Gavi has funded in country  $i$  prior to year  $t$  that can impact cause of death  $c$ . We discuss the connections between vaccines and causes of death in Section IID.

In specifications (3) and (5), we estimate the immediate impact of Gavi funding on child mortality even though not all children under 5 will be vaccinated immediately, as most Gavi funding focuses on routine immunization of infants. We expect the short-run effect of vaccine introduction on child mortality to be smaller than the long-run effect, as it will take years for all children under 5 to be vaccinated. The  $\beta_1$  coefficient in specifications (3) and (5) will average these short-run and long-run effects. However, the event-study specification will allow us to separately estimate the short-run and longer-run effects.<sup>10</sup>

A recent literature has documented bias in standard DD estimators in the presence of differential timing of treatment and heterogeneous treatment effects (see, e.g., De Chaisemartin and d'Haultfœuille 2020; Goodman-Bacon 2021; Sun and Abraham 2021; Callaway and Sant'Anna 2021). To ensure that our results are robust to correcting for these biases, we rely on the methods of BJS. Their estimator imputes counterfactual outcomes for treated observations, using an econometric model estimated only on the nontreated observations (including never-treated countries). Observation-level treatment effects are calculated by differencing the actual and counterfactual outcomes. Then, overall treatment effects are calculated by averaging over these observation-level treatment effects. The BJS estimator is robust to treatment effect heterogeneity and is finite-sample efficient. We implement the estimator using their provided Stata code.

### B. Variation in Gavi Funding

We obtained a list of launch dates of Gavi funding for each supported vaccine in each country from Gavi (2019). When multiple launch dates were listed, we used

<sup>10</sup> Most of Gavi's support funds routine immunization of infants, but Gavi occasionally funds catch-up campaigns (for MR, for example) for which older children are eligible. Theoretically, our child mortality results will include the impact of such campaigns since we examine child mortality, not infant mortality, but our coverage estimates would be biased toward zero because our control cohorts, born before Gavi introduced funding, would be partially treated. In practice, measles, rubella, and polio catch-up campaigns are unlikely to affect our results, since funds were not specifically allocated for these campaigns until 2016, the last year for which we have coverage data. Funding for meningitis A campaigns began in 2011, but we do not have coverage data for meningitis, and campaigns for yellow fever began in 2013, but yellow fever coverage accounts for very few of our observations.

TABLE 1—NUMBER OF COUNTRIES INTRODUCING A GAVI-FUNDED VACCINE, BY YEAR AND VACCINE

| Year     | Inactivated polio vaccine | Japanese encephalitis | Measles or MR | Meningitis A | Pentavalent (diphtheria, tetanus, pertussis, hepatitis B, Haemophilus influenzae type B) | Pneumococcal | Rotavirus | Yellow fever |
|----------|---------------------------|-----------------------|---------------|--------------|------------------------------------------------------------------------------------------|--------------|-----------|--------------|
| Pre-2003 |                           |                       |               |              | 11                                                                                       |              |           | 7            |
| 2003     |                           |                       |               |              |                                                                                          |              |           | 3            |
| 2004     |                           |                       |               |              |                                                                                          |              |           | 4            |
| 2005     |                           |                       |               |              | 6                                                                                        |              |           | 2            |
| 2006     | 1                         |                       |               |              | 3                                                                                        |              |           |              |
| 2007     |                           |                       | 1             |              | 4                                                                                        |              |           | 1            |
| 2008     |                           |                       | 1             |              | 18                                                                                       |              | 1         | 2            |
| 2009     |                           |                       |               |              | 17                                                                                       | 2            | 1         |              |
| 2010     |                           |                       |               | 3            | 3                                                                                        | 1            | 2         |              |
| 2011     |                           |                       | 1             | 3            | 3                                                                                        | 13           | 1         | 1            |
| 2012     |                           |                       | 8             | 4            | 5                                                                                        | 8            | 7         |              |
| 2013     |                           |                       | 10            | 2            | 2                                                                                        | 14           | 6         |              |
| 2014     | 3                         |                       | 11            | 3            | 1                                                                                        | 8            | 16        | 1            |
| 2015     | 39                        | 1                     | 10            | 1            |                                                                                          | 8            | 4         |              |
| 2016     | 14                        | 2                     | 5             | 3            |                                                                                          | 3            | 3         |              |
| 2017     | 2                         | 1                     | 6             | 2            |                                                                                          | 1            | 3         |              |
| 2018     | 15                        | 1                     | 8             | 1            |                                                                                          | 1            | 2         |              |
| 2019     | 2                         |                       | 6             | 1            |                                                                                          | 1            |           |              |
| Total    | 76                        | 5                     | 67            | 23           | 73                                                                                       | 60           | 46        | 21           |

*Notes:* This table lists the number of countries that introduced, with Gavi support, each vaccine in each year. In the interest of space, this table does not include introductions of the human papillomavirus vaccine, and groups together measles and MR. The complete table is available in the Supplemental Appendix.

*Source:* Gavi

the earliest date to minimize concerns about endogeneity.<sup>11</sup> We also obtained, from Gavi's website, data on the amount of funds approved to be disbursed to each country each year under various subcategories (Gavi 2020). Some subcategories indicate the vaccine for which the money is approved, while some subcategories are broader (such as "Health Systems Strengthening"). We define the first year that Gavi funded a vaccine as the first year that money is approved for a vaccine-specific line item. We primarily use the launch dates but find similar results using the financial data.<sup>12</sup>

Table 1 presents the number of countries that first introduced, with Gavi support, a particular vaccine in each year using the launch dates. From the table we can see that each vaccine was introduced in countries around the world in a staggered manner and that the timeline differs for each vaccine (e.g., both the pneumococcal and rotavirus vaccines are first introduced around 2008–2009, but the pneumococcal vaccine was introduced in more countries and on average earlier). Some vaccines were more targeted (Gavi launched the Japanese encephalitis vaccine in only five countries), while others were distributed more broadly. Supplemental Appendix Table A1 presents this information using the first year a country received Gavi funding in a vaccine-specific subcategory. Supplemental Appendix Table A2 lists the countries in our sample and which of these countries have ever received Gavi funding.

<sup>11</sup> Multiple launch dates often coincide with funding for follow-up campaigns to help vaccinate older children.

<sup>12</sup> Allowing for errors where the launch date is one year before or after the first year of funding, these two data sources match for 92 percent of country-vaccine pairs.

This staggered introduction results in substantial within-country variation in when each vaccine was launched with Gavi support, which we exploit for identification. For example, Angola introduced the pentavalent vaccine with Gavi funding in 2006, but did not introduce the pneumococcal vaccine until 2013, the rotavirus vaccine until 2014, and the MR vaccine until 2018. Supplemental Appendix Figure A1 maps the geographic variation in the timing of Gavi's expansion of funding for three important vaccines: pentavalent, pneumococcal, and rotavirus using shapefiles from the World Bank (2023).

Our identification strategy rests on the exogeneity of the timing of Gavi funding for a given vaccine in a given country and the extent to which this timing is unrelated to other vaccine- or disease-specific changes. For example, if timing is externally driven based on Gavi's priorities or global supply constraints, our identifying assumptions would hold. Even locally driven determinants of Gavi funding would not bias our estimates as long as they are uncorrelated with other factors that affect vaccine- or disease-specific outcomes. In the rest of this subsection, we discuss the myriad of determinants of country adoption of Gavi-sponsored vaccines and a robustness check that relies solely on the externally driven variation.

The factors that lead countries to adopt certain vaccines, with or without Gavi support, at certain times are varied. Burchett et al. (2012) stresses the importance of funding availability, political will to promote vaccines, and the local disease burden of the various pathogens. Similarly, Makinen et al. (2012) argues that countries are influenced by WHO recommendations, the availability of country-specific data on the burden of disease, and the overall affordability of the vaccine.

In the context of Gavi-supported introduction, we identified several factors that have driven vaccine adoption: pricing and sustainability concerns, local vaccine infrastructure, specific Gavi recommendations and policies, the availability of local burden of disease estimates, WHO recommendations, and vaccine supply constraints. For example, according to Gavi (2016), the rollout of the rotavirus vaccine did not meet the initial coverage targets by 2015 due to affordability concerns. Affordability is a major factor, as countries are typically committing to perennial purchases of the vaccine, as removing vaccines from a national immunization program is a rare occurrence. In addition, discounted vaccine prices are guaranteed only for a limited number of years. The same report notes that concern about cold chain quality also slowed uptake.

Similarly, Gavi policy required pentavalent coverage above 70 percent to apply for some new vaccine support, but did not apply this threshold to the Japanese encephalitis and meningitis A vaccines. When the pneumococcal vaccine was introduced, countries had to have achieved 70 percent or higher coverage of DTP to apply for Gavi support (Gavi 2018). Gavi also discouraged some countries from applying for new pentavalent support due to an expected global shortage of the vaccine (Chee et al. 2008). The availability of a burden of disease report to make the case that vaccine purchase is cost-effective is an additional factor that drives adoption. Initially, Gavi recommended a burden of disease study before a country applied for Hib support, for example (Chee et al. 2008), and the WHO did not initially recommend rotavirus vaccine outside of the Americas and Europe because of the absence of such a study (WHO 2007).

WHO recommendations through its Strategic Advisory Group of Experts also influence take-up. For example, a 2014 recommendation to improve coverage of the meningococcal A vaccine was followed by increased vaccine adoption in the meningitis belt, which stretches from Senegal to Ethiopia in sub-Saharan Africa (WHO 2019). There is regional variation as well in WHO recommendations. In 2013, 11 member states of the South East Asian Region of the WHO endorsed a resolution to promote the rubella vaccine. Subsequently, Bangladesh, India, Myanmar, and Indonesia introduced the MR vaccine (WHO 2013). Finally, our personal correspondence with Gavi indicated that the distribution of both the pentavalent and pneumococcal vaccines was significantly constrained by manufacturing supply capacity, leading to a staggered rollout of the vaccine across countries.

Overall, while much variation in the timing of Gavi support for a given vaccine in a given country is externally driven, there is the possibility of locally driven variation as well. Since countries have to apply for Gavi funding for a vaccine and commit to Gavi's cost-sharing agreement, the Gavi-funded introduction of a vaccine could signal how a country is prioritizing different vaccines. In other words, a country may apply for Gavi support for a vaccine while also planning a non-Gavi-related push to increase coverage of that specific vaccine. Our *country*  $\times$  *cohort* fixed effects control for any non-vaccine-specific changes in a country's public health or immunization policies, and our *vaccine*  $\times$  *cohort* fixed effects control for vaccine-specific (global) changes, but these fixed effects do not eliminate bias from a country- and vaccine-specific change in policy.

In our view, it is unlikely that such a change would coincide perfectly with Gavi support, and therefore should result in evidence of preexisting differential trends, which we test for and do not see. However, we address the possibility of a country-disease-specific initiative in a robustness check with an instrumental variable that is exogenous to recipient country policy. Specifically, we use country eligibility (based on Gavi's criteria determined in 1999) interacted with global Gavi funding for a given vaccine in a given year (minus own country receipts). This instrument relies on the exogenous, external sources of variation such as vaccine supply constraints and changes in international priorities discussed above and is independent of recipient country policy changes.

### C. Vaccine Coverage Data

We obtained data on vaccine coverage across countries, vaccines, and cohorts from the WHO and UNICEF (World Health Organization n.d.). The WHO makes available both survey estimates of vaccine coverage and official estimates that combine data from administrative sources and survey sources.<sup>13</sup> Due to concerns about the accuracy of the administrative data underpinning the official estimates amid

<sup>13</sup> The official estimates are taken from administrative data with some editing to account for the possibility of inaccurate data. For example, the WHO/UNICEF replaces administrative data if coverage rates calculated from survey data strongly indicate discrepancies. The WHO/UNICEF also imputes values for missing data using linear interpolation. Other corrections are made to reconcile inconsistencies between dose numbers.

TABLE 2—SUMMARY STATISTICS ON VACCINE COVERAGE

| Vaccine                        | Mean coverage |         |         |         |         |         |
|--------------------------------|---------------|---------|---------|---------|---------|---------|
|                                | 1985–89       | 1990–94 | 1995–99 | 2000–04 | 2005–09 | 2010–14 |
| BCG vaccine                    | 76            | 83      | 82      | 84      | 88      | 91      |
| DTP vaccine                    | 69            | 75      | 74      | 78      | 81      | 84      |
| Hepatitis B vaccine            |               | 75      | 81      | 82      | 77      | 83      |
| HiB vaccine                    |               |         | 96      | 81      | 80      | 84      |
| Inactivated polio vaccine      |               |         |         |         |         | 84      |
| Measles containing vaccine     | 60            | 68      | 69      | 74      | 75      | 78      |
| Pneumococcal conjugate vaccine |               |         |         |         | 59      | 69      |
| Polio vaccine                  | 69            | 75      | 75      | 80      | 82      | 84      |
| Rubella containing vaccine     |               |         |         |         | 87      | 85      |
| Rotavirus vaccine              |               |         |         |         | 60      | 69      |
| Tetanus vaccine                |               |         |         | 40      |         | 57      |
| Yellow fever vaccine           | 33            | 36      | 29      | 56      | 63      | 68      |

*Notes:* This table presents average coverage rates by vaccine for the countries and cohorts for which WHO/UNICEF-collected survey data are available. Summary statistics are only provided for cells with at least two observations. Observations are at the *country × vaccine × dose × cohort* level. Our data include estimates of coverage for cohorts born from 1979 to 2016, but 96 percent of the observations are from 1985 to 2014.

incentives to inflate these data (Lim et al. 2008; Sandefur and Glassman 2015), we use the survey data for our main estimates.<sup>14</sup>

The WHO and UNICEF collected data from more than 881 surveys across 152 countries to calculate coverage rates of various vaccines for different cohorts. For example, the data include coverage estimates for 16 different vaccine doses (separating out multiple doses of some vaccines) for the cohort born in 2012 in Ghana as reported in the Ghana Demographic and Health Survey in 2014. Data for the United States, another illustrative example, comes from the National Immunization Surveys from 2002 to 2014 (although the United States is not included when we restrict the sample based on country income). Some data are based on immunization cards viewed by enumerators, while some are reported by household members.<sup>15</sup>

Table 2 reports summary statistics from these data on 12 different vaccines from 1985 to 2014. Since surveys are conducted at varying intervals and usually only collect data on coverage for recently born cohorts, there are missing observations. On average, we have coverage rates of any vaccine for eight cohorts per country, and for each country-cohort with data, we have coverage rates for seven vaccines (counting each dose separately). For most vaccines, we see increases in coverage over time, but these are likely to underestimate the true growth as the sample is skewed toward more-developed countries earlier in the time period. Gavi has provided funding for all the vaccines listed in Table 2 in some countries in some years, except for the BCG vaccine and the conventional polio vaccine. Supplemental Appendix Table A2 provides

<sup>14</sup> Gavi creates one incentive to inflate official estimates since the introduction of new vaccines is predicated on reaching a coverage threshold for DTP, as described above.

<sup>15</sup> One caveat when using survey data is that the immunization status of children who have died is not recorded (McGovern and Canning 2015). Our reliance on within-country, cross-vaccine variation helps us with this problem. In order to bias our results, the differential coverage rate for a funded vaccine relative to an unfunded vaccine would have to differ for children who died relative to children who survived. Specifically, in order to negate our result, children who died would have to be *less* likely to receive the Gavi-funded vaccines than other vaccines relative to children who survived. This seems unlikely, but we also note that the official estimates, which are primarily based on administrative data collected from service providers, give us similar results.

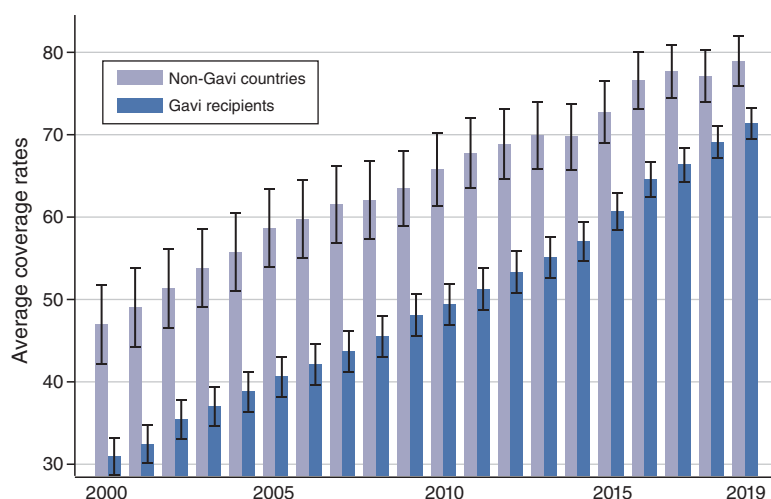


FIGURE 1. VACCINE COVERAGE OVER TIME

*Note:* This figure plots average coverage rates across all vaccines, separately for Gavi recipients and other LMI countries with 2000 GNI/GDP per capita < 3,500 (2010 US dollars), along with 95 percent confidence intervals. The data are from the WHO/UNICEF estimates of national immunization coverage, and missing values in years before the first nonmissing value have been imputed as 0 to create a balanced sample.

more data details, including, for each country, whether it ever received Gavi funding and whether it is included when we restrict the sample based on income. The table also lists the first and last year for which we have coverage data for each vaccine and each country. Note that surveys are not conducted in every year, so we do not have coverage data for every year in these ranges.<sup>16</sup>

To illustrate how the data drive our regression results, we provide a descriptive analysis of vaccine coverage using the WHO/UNICEF official estimates.<sup>17</sup> Figure 1 and Figure 2 illustrate the increase in vaccine coverage over time for Gavi recipients (dark blue) and non-Gavi lower-income countries (light gray) since Gavi's founding for all vaccines (Figure 1) and for the new vaccines (Figure 2). Demonstrating the cross-country variation that we exploit, these descriptive statistics suggest that Gavi helped close gaps in coverage of preexisting vaccines and, in fact, helped recipients reach higher coverage of newer vaccines than similar countries. Next, Figure 3 demonstrates the cross-vaccine variation we use, presenting trends in vaccine coverage for preexisting and new vaccines separately among Gavi recipients. While both types of vaccines are eligible for Gavi support, the dramatic increase in coverage for new vaccines is striking. Finally, we note that

<sup>16</sup> Not surprisingly, there are fewer years of data available for the newer vaccines, pneumococcal and rotavirus, but these years coincide with Gavi funding. Thus, even though Angola, for example, only has data on the new vaccines for 2013 and 2014, this allows us to observe coverage both prior to and after Gavi introduction of the pneumococcal vaccine. Moreover, while survey data are sometimes sparse, we find similar results with the WHO official estimates of coverage.

<sup>17</sup> These figures do not use the survey data, since it is not a balanced sample, making graphs of raw data difficult to interpret. We also impute missing values in years before the first nonmissing value for a given country and given vaccine as 0 since there were no official estimates of newer vaccines before the vaccines were available.



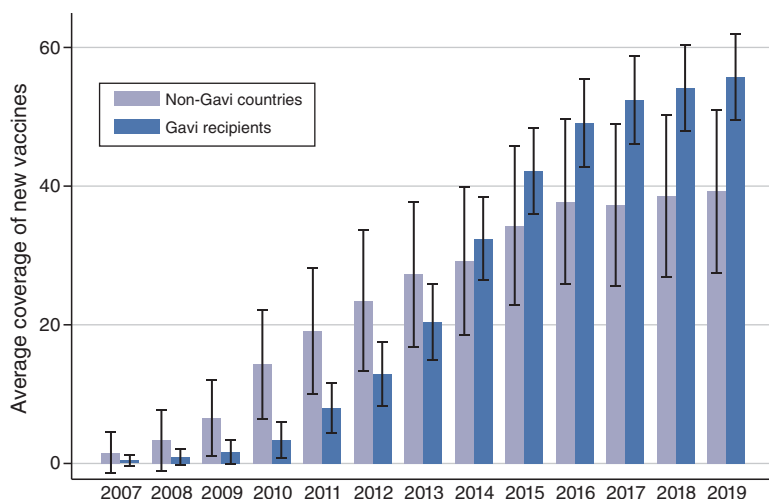


FIGURE 2. VACCINE COVERAGE OVER TIME FOR THE PNEUMOCOCCAL AND ROTAVIRUS VACCINES

*Notes:* This figure plots average coverage rates of the pneumococcal and rotavirus vaccines, separately for Gavi recipients and other LMI countries with 2000 GNI/GDP per capita < 3,500 (2010 US dollars), along with 95 percent confidence intervals. The data are from the WHO/UNICEF estimates of national immunization coverage, and missing values in years before the first nonmissing value have been imputed as 0 to create a balanced sample.

the preexisting cross-country differences in coverage in Figures 1 and 2 and the preexisting cross-vaccine differences in Figure 3 help motivate our triple-difference strategy where we can control for time-varying differences across countries and vaccines.

#### D. Child Mortality Data

We obtained data on child mortality rates by cause from Perin et al. (2022b), who provide the most recent update of the WHO's GHO data. These data give the number of deaths per 1,000 live births from groups of causes for each year from 2000 to 2019 and for almost 200 countries. The data are separated into deaths among 0–27-day-old infants and children between 1 and 59 months of age. We focus on 1–59-month mortality (hereafter, *child mortality*), since newborns are not generally vaccinated in the first month of life (except for the Hepatitis B birth dose and BCG, neither of which was supported by Gavi during this time period). Our results are also robust to using all under-5 deaths.<sup>18</sup>

The methods used to construct these mortality data are described in Perin et al. (2022a). When available, the authors use vital registration systems documenting cause of death to estimate the number of deaths attributable to each cause. However, for some countries, especially LMI countries, high-quality vital registration data are not available. In these cases, the authors rely on verbal autopsy studies (surveys

<sup>18</sup> One might think that newborns could be used as a placebo group; however, they would likely be protected via spillovers if older children are vaccinated.

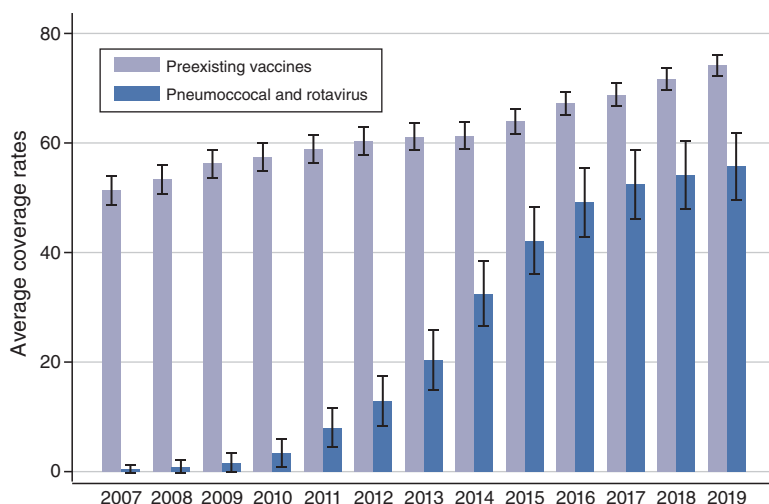


FIGURE 3. VACCINE COVERAGE OVER TIME FOR GAVI RECIPIENTS, BY GROUPS OF VACCINES

*Notes:* This figure plots average coverage rates of preexisting vaccines (BCG, DTP, HepB, Hib, inactivated polio vaccine, measles, conventional polio vaccine, rubella, and yellow fever) and two new vaccines (pneumococcal and rotavirus), separately. The sample is restricted to Gavi recipients. The data are from the WHO/UNICEF estimates of national immunization coverage, and missing values in years before the first nonmissing value have been imputed as 0 to create a balanced sample.

of close survivors that are used by trained professionals to attribute cause of death) to estimate mortality-by-cause data. Since these studies are not available for every country and year, the authors also use statistical models to impute mortality-by-cause data. Since our empirical strategy relies on variation in mortality by cause, errors in the attribution of deaths to specific causes, due to either imputation or human error, may affect our results. If these errors are unrelated to Gavi funding, they would simply lead to attenuation bias. However, some of these statistical models include data on vaccination rates.<sup>19</sup> Therefore, to ensure that our conclusions are not driven by these imputations, we use the mortality estimates based solely on VR data, which do not include any imputations for our main results.

Supplemental Appendix Table A2 lists which countries have data in the VR subset of the GHO data. Supplemental Appendix Figure A2 displays this information on a map. There are 19 Gavi recipients who have any VR data, all of which are currently middle-income countries (some upper middle-income, but some lower middle-income as well). Since this is a selected sample of countries, we provide robustness checks using all of the GHO data as well as data from the Global Burden of Disease Study (GBD) (Vos et al. 2020; Global Burden of Disease Collaborative Network 2019). While the GBD also relies on both vital registration and verbal

<sup>19</sup> Specifically, the authors fit a multinomial logit model to verbal autopsy data from at least 130 studies across more than 41 countries. They rely only on higher-quality studies, where their inclusion criteria include the following, for example: the study must have been conducted in 1980 or later, have more than one cause of death, and examine more than 25 deaths, and less than 25 percent of deaths can be attributed to unknown causes. Moreover, the authors exclude studies that do not use standardized questionnaires or methods.

TABLE 3—SUMMARY STATISTICS ON MORTALITY BY CAUSE

| Cause of death                                        | Mean deaths per 1,000 live births |             |                 |             |                              |             |
|-------------------------------------------------------|-----------------------------------|-------------|-----------------|-------------|------------------------------|-------------|
|                                                       | All countries                     |             | Gavi recipients |             | Gavi recipients with VR data |             |
|                                                       | 2000<br>(1)                       | 2019<br>(2) | 2000<br>(3)     | 2019<br>(4) | 2000<br>(5)                  | 2019<br>(6) |
| HIV/AIDS                                              | 2.7                               | 0.5         | 4.4             | 0.9         | 0.2                          | 0.1         |
| Diarrheal diseases                                    | 6                                 | 2           | 13              | 4.2         | 2.5                          | 0.4         |
| Tetanus                                               | 0                                 | 0           | 0               | 0           | 0                            | 0           |
| Measles                                               | 2.7                               | 0.8         | 6.1             | 1.7         | 0                            | 0.1         |
| Meningitis/encephalitis                               | 1.3                               | 0.4         | 2.8             | 0.7         | 0.4                          | 0.2         |
| Malaria                                               | 5.5                               | 1.8         | 13.4            | 4.4         | 0.1                          | 0.2         |
| Lower respiratory infections                          | 7.4                               | 3           | 15.1            | 5.9         | 3.8                          | 1.2         |
| Prematurity                                           | 0.5                               | 0.4         | 0.6             | 0.6         | 0.1                          | 0.1         |
| Birth asphyxia and birth trauma                       | 0.3                               | 0.2         | 0.4             | 0.4         | 0.2                          | 0.1         |
| Sepsis and other infectious conditions of the newborn | 0                                 | 0           | 0               | 0           | 0                            | 0           |
| Other group 1 (postneonatal only)                     | 3.7                               | 1.7         | 7.7             | 3.7         | 0.8                          | 0.3         |
| Congenital anomalies                                  | 1.5                               | 1.3         | 1.2             | 1.4         | 1.7                          | 1.4         |
| Other noncommunicable diseases                        | 2.1                               | 1.3         | 3.1             | 1.7         | 1.5                          | 1.1         |
| Injuries                                              | 1.4                               | 1.4         | 2               | 2.4         | 1.9                          | 1           |
| Tuberculosis                                          | 0.6                               | 0.5         | 1.4             | 0.9         | 0.2                          | 0.4         |

*Note:* This table presents the average number of deaths among children 1–59 months old per 1,000 live births attributed to specific causes in each year for all countries, Gavi recipient countries, and Gavi recipient countries with VR data separately.

autopsy data, it uses different statistical models to impute mortality-by-cause data and allows us to ensure that our full sample results are not driven by the modeling choices specific to Perin et al. (2022a). Unfortunately, the GBD data do not identify which estimates are derived from VR data and which are imputed. Finally, we show that our results are robust to alternate specifications that do not rely as precisely on cause-of-death attribution by comparing the effect of vaccine introduction on all vaccine-preventable causes of death relative to all other causes.

Table 3 lists the 15 causes of death available in the GHO data and provides means from 2000 and 2019 for all countries, Gavi recipients, and for Gavi recipients with VR data.<sup>20</sup> Mortality rates have generally fallen over time for almost all causes, but they vary substantially across causes, as do the rates of decline. For example, lower respiratory infections accounted for the highest number of child deaths, at almost 8 per 1,000 live births in 2000, and the rate fell by more than 50 percent from 2000 to 2019. Tuberculosis accounted for a much lower number of deaths, at 0.6 per 1,000 live births in 2000, and the rate fell by 29 percent from 2000 to 2019. Mortality rates are higher in Gavi countries; they are more than twice as high for respiratory infection and diarrheal deaths.

As discussed above in Section IIA, we assign the vaccines that Gavi funds to these causes of death in multiple ways since some vaccines can prevent deaths attributed to multiple causes. These linkages are based on the WHO's (2015) *Immunization in Practice: A Practical Guide for Health Staff* and are in consultation with a public

<sup>20</sup> We omit "Sepsis and other infectious conditions of the newborn" and "Tetanus" from our analysis since they account for nearly zero deaths among children aged 1–59 months.

TABLE 4—LINKS BETWEEN CAUSES OF DEATH AND VACCINES

| Cause of death               | Primarily linked<br>Gavi-funded vaccine<br>(1) | Any linked<br>Gavi-funded vaccines<br>(2)                         |
|------------------------------|------------------------------------------------|-------------------------------------------------------------------|
| Diarrheal diseases           | Rotavirus                                      | Rotavirus, measles                                                |
| Measles                      | Measles, MR                                    | Measles, MR                                                       |
| Meningitis/encephalitis      | Meningitis A, Japanese encephalitis            | Meningitis A, Japanese encephalitis,<br>pneumococcal, pentavalent |
| Lower respiratory infections | Pneumococcal, Hib (pentavalent)                | Pneumococcal, Hib (pentavalent), measles                          |

Notes: Column 1 assigns vaccines to the cause of death to which they are most closely linked. Column 2 lists any vaccine that can impact deaths attributed to each cause.

health expert, Richard A. Cash, MD, MPH, and senior lecturer at the Harvard T. H. Chan School of Public Health. When estimating specifications (3) and (4), we maximize the within-country variation across causes of death by focusing on the cause of death most closely linked to each vaccine. These causes are indicated in column 1 of Table 4. Specifically, the measles, meningitis, and Japanese encephalitis vaccines are linked to their eponymous causes of death, the rotavirus vaccine is linked to diarrheal deaths, and the pneumococcal and Hib parts of the pentavalent vaccines are linked to respiratory deaths. On the other hand, specification (5) allows vaccines to affect multiple causes of death. Some diseases, such as measles, can compromise immunity; the measles vaccine could reduce the number of child deaths attributed to diarrheal diseases or respiratory infections. In addition, the pneumococcal and pentavalent vaccine can prevent diseases that, in some rare cases, lead to meningitis. Column 2 of Table 4 indicates the vaccine–cause-of-death linkages we use to calculate the number of vaccines that Gavi has introduced that could affect each cause of death. Specification (5) is, in some ways, more complete, allowing all vaccines to affect the causes of death that might result from the disease prevented. However, meningitis is a very rare consequence of pneumococcal disease, for example, and it is unrealistic to expect the pneumococcal vaccine to affect meningitis deaths as much as it affects pneumonia deaths. Hence, while we estimate both specifications (3) and (5), we prefer specification (3). If the measles vaccine, for example, also reduces deaths from other causes, that would only bias our estimates from specification (3) toward zero since we are comparing measles deaths to other deaths after Gavi funded the measles vaccine. Focusing on the most affected cause of death in specification (3) also allows us to estimate the event study, specification (4).

### III. Results

#### A. Impact of Gavi Funding on Vaccine Coverage

Table 5 presents estimates of the impact of the Gavi-funded introduction of a vaccine on the coverage rate, using specification (1) and the WHO/UNICEF-compiled survey data on vaccine coverage. Panel A estimates the regression with data on all countries in the dataset, while panels B and C limit the sample to countries with GNI per capita in 2000 below \$10,000 in 2010 dollars and below \$3,500 in 2010

TABLE 5—IMPACT OF GAVI-FUNDED INTRODUCTION OF A VACCINE ON COVERAGE RATE

| Estimation method                                                                       | OLS          |              |                            |                                  | BJS          |              |                            |                                  |
|-----------------------------------------------------------------------------------------|--------------|--------------|----------------------------|----------------------------------|--------------|--------------|----------------------------|----------------------------------|
|                                                                                         | All          | Preexisting  | Pneumococcal and rotavirus | Pneumococcal, rotavirus, and HiB | All          | Preexisting  | Pneumococcal and rotavirus | Pneumococcal, rotavirus, and HiB |
| Funded vaccines                                                                         | (1)          | (2)          | (3)                        | (4)                              | (5)          | (6)          | (7)                        | (8)                              |
| <i>Panel A. All countries</i>                                                           |              |              |                            |                                  |              |              |                            |                                  |
| Postintroduction                                                                        | 3.4<br>(0.9) | 2.3<br>(0.8) | 19.9<br>(7.5)              | 11.1<br>(4.8)                    | 4.8<br>(1.0) | 4.5<br>(1.1) | 17.8<br>(6.5)              | 14.5<br>(4.9)                    |
| Observations                                                                            | 8,641        | 8,528        | 7,364                      | 7,702                            | 7,810        | 7,793        | 7,130                      | 7,136                            |
| R <sup>2</sup>                                                                          | 0.83         | 0.83         | 0.84                       | 0.84                             |              |              |                            |                                  |
| Dep. var. mean                                                                          | 76.9         | 76.9         | 76.3                       | 76.6                             | 77.1         | 77.1         | 76.7                       | 76.7                             |
| <i>Panel B. Gavi recipient or 2000 GNI/GDP per capita &lt; 10,000 (2010 US dollars)</i> |              |              |                            |                                  |              |              |                            |                                  |
| Postintroduction                                                                        | 2.9<br>(0.9) | 2.1<br>(0.8) | 15.2<br>(8.4)              | 7.9<br>(4.9)                     | 4.7<br>(1.1) | 4.3<br>(1.1) | 18.5<br>(6.7)              | 13.3<br>(4.9)                    |
| Observations                                                                            | 8,087        | 7,973        | 6,812                      | 7,148                            | 7,240        | 7,223        | 6,560                      | 6,566                            |
| R <sup>2</sup>                                                                          | 0.82         | 0.82         | 0.83                       | 0.83                             |              |              |                            |                                  |
| Dep. var. mean                                                                          | 76.1         | 76.0         | 75.3                       | 75.6                             | 76.2         | 76.2         | 75.6                       | 75.6                             |
| <i>Panel C. Gavi recipient or 2000 GNI/GDP per capita &lt; 3,500 (2010 US dollars)</i>  |              |              |                            |                                  |              |              |                            |                                  |
| Postintroduction                                                                        | 2.7<br>(0.9) | 2.3<br>(0.8) | 10.2<br>(8.3)              | 5.0<br>(4.6)                     |              | 4.7<br>(1.2) |                            |                                  |
| Observations                                                                            | 6,997        | 6,884        | 5,723                      | 6,058                            |              | 6,068        |                            |                                  |
| R <sup>2</sup>                                                                          | 0.83         | 0.84         | 0.84                       | 0.84                             |              |              |                            |                                  |
| Dep. var. mean                                                                          | 75.5         | 75.5         | 74.4                       | 74.9                             |              | 75.6         |                            |                                  |

*Notes:* This table presents estimates of the impact of the Gavi-funded introduction of a vaccine on coverage rates using the WHO/UNICEF-collected survey data. Observations are at the *country*  $\times$  *vaccine*  $\times$  *dose*  $\times$  *cohort* level. Columns 1–4 provide OLS estimates while columns 5–8 provide estimates using the method in BJS. All regressions include *country*  $\times$  *vaccine*, *country*  $\times$  *cohort*, and *vaccine*  $\times$  *cohort* fixed effects. Panel A includes data on 147 countries; panel B includes only the 124 countries with GNI/GDP per capita in 2000 < 10,000 (2010 US dollars); and panel C includes the 101 countries with GNI/GDP per capita in 2000 < 3,500 (2010 US dollars). Preexisting vaccines (columns 2 and 6) refer to all vaccines except pneumococcal and rotavirus. Robust standard errors, clustered by country, are shown in parentheses.

dollars, respectively. Countries that received Gavi funding for the introduction of any vaccine are included in all samples given Gavi's eligibility criteria.<sup>21</sup> All specifications include *country*  $\times$  *vaccine*, *country*  $\times$  *cohort*, and *vaccine*  $\times$  *cohort* fixed effects. Columns 1 to 4 use standard OLS methods to estimate the impacts, while columns 5 to 8 use the BJS method to address the bias arising from heterogeneous effects and staggered treatment. Columns 1 and 5 include coverage data from all vaccines. Columns 2 and 6 examine Gavi funding of previously available vaccines, while columns 3 and 7 focus on Gavi funding of new vaccines (pneumococcal and

<sup>21</sup> Gavi eligibility is based on GNI per capita in 1998; however, we use GNI per capita in 2000 because it is available for a larger number of countries (World Bank 2020). Even still, GNI per capita in 2000 is not available for all countries. All Gavi-eligible countries with data in 2000 had GNI per capita of less than \$1,860 (in 2010 dollars) except for Cuba, which had GNI per capita of \$3,411 (Cuba received Gavi eligibility because of the lack of reliable data on Cuban GDP in 2000). Therefore, we choose \$3,500 as our most restrictive threshold, adding 17 countries that never received Gavi funding. While our results for preexisting vaccines are not sensitive to the income thresholds, the stricter income cutoff limits our statistical power when it comes to estimating impacts for newer vaccines due to limited coverage data prior to Gavi funding. Thus, we include the more expansive set of countries with GNI per capita < \$10,000 (2010 US dollars) to eliminate the highest-income countries while retaining a sufficient sample to estimate effects for the newer vaccines.

rotavirus). To be consistent with our mortality results below, we also include columns 4 and 8, grouping Hib with the new vaccines. Recall that BJS methods provide observation-level treatment effects; thus, in columns 6 to 8, we simply average the treatment effects for the relevant vaccines. When using OLS, we estimate different effects for preexisting and new vaccines by including observations on all vaccines before they are funded in a particular country, but including postlaunch observations only on the column-specific vaccines.<sup>22</sup>

We see a significant 2–3 percentage point increase in coverage after a vaccine is introduced with Gavi assistance (column 1). This muted result may be expected given the high existing coverage rates of some of these vaccines prior to the availability of Gavi funding. Along these lines, we see a much larger impact for the newly available vaccines (column 3) relative to vaccines that had preexisting distribution (column 2). The coverage rates for pneumococcal and rotavirus increased by 10–20 percentage points, depending on the choice of control countries. Separate estimates for each treated vaccine support our conclusion that the largest increases in coverage were for pneumococcal and rotavirus and also suggest that the effects on preexisting vaccines are driven by the pentavalent vaccine (DTP, HepB, and Hib), one of the vaccines that Gavi promoted early on to modernize vaccine programs in developing countries (Supplemental Appendix Table A3).

Using BJS methods (columns 5–8), we find larger point estimates for all vaccines, closer to 5 percentage points, which is to be expected since these methods do not use previously treated country-vaccines as a control for the countries that receive those vaccines later in the time period.<sup>23</sup> Given limited availability of pretreatment coverage data for the newer vaccines in the lowest-income countries, we are unable to estimate the BJS regression in panel C, column 7–8, but find results very similar to our baseline results using the set of countries with incomes less than \$10,000 (panel B).<sup>24</sup>

As discussed above, the full set of two-way interacted fixed effects account for a host of possible omitted variables. The identifying assumption is that coverage of Gavi-funded vaccines would have trended similarly to coverage of non-Gavi-funded vaccines, in adopting and nonadopting countries, in the absence of any Gavi funding. To provide support for this assumption, we look for differences in trends

<sup>22</sup> Limiting the sample entirely to observations for preexisting vaccines gives us very similar results. Doing the same for newer vaccines gives us qualitatively similar results that are often less precisely estimated because we have very few observations of pretreatment coverage, although, for the strictest income threshold, the point estimate is of similar magnitude and significant at the 10 percent level.

<sup>23</sup> Our results are not inconsistent with Dykstra et al. (2019). They find that Gavi increased coverage of preexisting vaccines by 4.7 percentage points, although the estimate is imprecise. We find effects between 2.2 and 4.5 percentage points. Dykstra et al.'s (2019) strategy of exploiting the discontinuity in income per capita that determined Gavi eligibility was not applicable for the newer vaccines and, for the preexisting vaccines, offers a local average treatment effect that applies to countries near the eligibility threshold. To explore how treatment effects vary with income, we exploit a feature of the BJS method that allows us to estimate treatment effects for each treated observation. We plot a nonparametric kernel regression of these treated effects for the preexisting vaccines on GDP per capita in 2000 (Supplemental Appendix Figure A3). The treatment effects rise with GDP per capita for the poorest countries among Gavi recipients, but fall back noisily to zero for the richest Gavi recipients (those closest to the eligibility threshold).

<sup>24</sup> BJS methods require that specific *country*  $\times$  *vaccine* combinations be observed in the pre-period to estimate treatment effects, whereas OLS relies on functional form assumptions in such cases. Panel C has many fewer never-treated countries than panel B, making it harder to use the BJS estimator. All our BJS estimates use a minimum effective number of treated observations of at least 16.



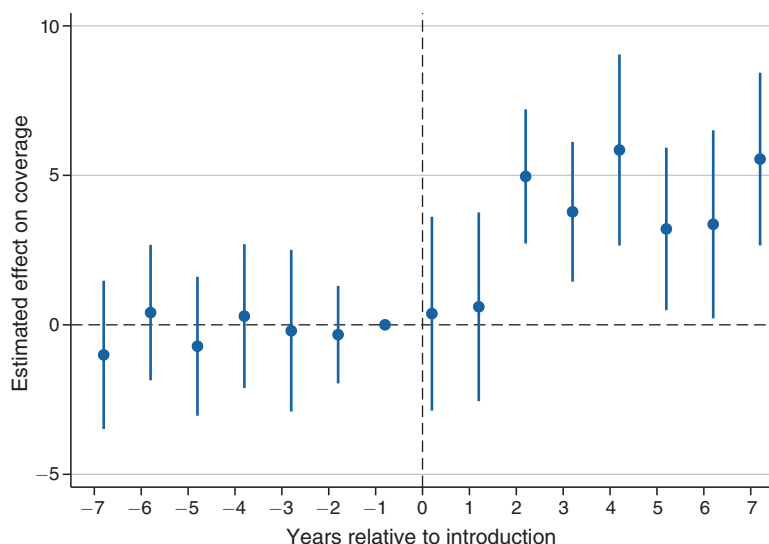


FIGURE 4. IMPACT OF GAVI-FUNDED INTRODUCTION OF A VACCINE ON COVERAGE RATE

*Notes:* This figure plots the impact of the Gavi-funded introduction of a vaccine on coverage rates using the WHO/UNICEF-collected survey data. Observations are at the *country*  $\times$  *vaccine*  $\times$  *dose*  $\times$  *cohort* level. The regression includes *country*  $\times$  *vaccine*, *country*  $\times$  *cohort*, and *vaccine*  $\times$  *cohort* fixed effects. The year before Gavi introduces a vaccine is the omitted year. Ninety-five percent confidence intervals are estimated from robust standard errors, clustered by country.

between these vaccines prior to the Gavi-funded introduction with our event-study specification (2). Figure 4 plots the conditional difference in coverage between Gavi-funded vaccines and other vaccines for each time period (relative to this difference one year prior to the introduction). Effects are plotted along with their 95 percent confidence intervals. Importantly, these effects cannot be driven by time trends or country-specific effects as we include cohort-specific country fixed effects.

Figure 4 supports our identifying assumption: it does not appear that coverage of these particular vaccines was trending differentially prior to Gavi introduction—an  $F$ -test of all preintroduction coefficients fails to reject the null hypothesis of no differences ( $p$ -value = 0.71). We also see a marked increase in coverage soon after Gavi funded a vaccine, beginning in year 2. An  $F$ -test of all postintroduction coefficients rejects the hypothesis of no effect ( $p$ -value < 0.01).

One benefit of the event-study specification is that it allows flexibility in the impact over time; we see that the increase in coverage rates occurs in year 2 and remains consistent over the next five or more years. A likely explanation for the lag in the impact on coverage is that the launch dates are announcement events and that vaccine distribution ramps up after that. Especially in large countries, Gavi's strategy based on WHO guidance is to stagger distribution and ramp up over a few years instead of aiming to distribute nationwide in the first year. Also contributing to this lag might be the fact that in the first year of Gavi-funded vaccine distribution, there are usually two cohorts being targeted—children born in the previous 12 months and newborns—and vaccines may not be available to vaccinate

both cohorts fully. Supplemental Appendix Figure A4 presents the event study using BJS's estimator; we see preexisting differences but no significant evidence of a trend prior to introduction as well as a greater average treatment effect postintroduction relative to preintroduction, consistent with the regression results in Table 5.<sup>25</sup>

Supplemental Appendix Figure A5 demonstrates the robustness of our coverage results to a number of variations on the data and specification (results are also presented in Supplemental Appendix Tables A4, A5, and A6), including (i) using vaccine-specific line items from funding data to determine the first year of Gavi funding; (ii) using the WHO/UNICEF official estimates of coverage instead of survey data; (iii) restricting the sample to Gavi recipients; (iv) excluding the oral polio vaccine and the hepatitis B birth dose, both of which are not supported by Gavi;<sup>26</sup> (v) excluding measles and rubella coverage to ensure that any correlations between efforts by Gavi and the Measles and Rubella Initiative, which also supports vaccination programs, are not driving our results;<sup>27</sup> (vi) including *country*  $\times$  *vaccine* linear trends; (vii) including *region*  $\times$  *vaccine*  $\times$  *cohort* fixed effects; (viii) restricting the sample to coverage of the final dose for each vaccine (e.g., the third dose of DTP); and (ix) including *country*  $\times$  *vaccine* *dose* and *cohort*  $\times$  *vaccine* *dose* fixed effects instead of *country*  $\times$  *vaccine* and *cohort*  $\times$  *vaccine* fixed effects. For the *region*  $\times$  *vaccine*  $\times$  *cohort* fixed effects (vii), we define regions loosely based on WHO regions except that we combine some regions together due to limited variation within smaller regions (i.e., splitting up Africa and Asia into two regions instead of three each). These fixed effects would account for any regional pushes toward eradicating certain diseases that might be correlated with Gavi funding.

### B. Impact of Gavi Funding on Under-5 Mortality from Related Causes

Increased vaccine coverage in LMI countries has long been a goal of global health policy, making the estimated impacts on vaccine coverage notable; however, we are ultimately interested in the effects of increased vaccine coverage on welfare. To this end, Table 6 presents estimates from specification (3) on child mortality (deaths per 1,000 live births) from vaccine-preventable causes using the launch dates to determine the first year of Gavi support and restricting the sample to the VR subset of the GHO data. Panel A includes data from all countries, while panels B and C limit the sample to Gavi-recipient countries and other lower-income countries. Columns 1–4 use standard OLS methods, while columns 5–8 use BJS's estimator. Columns 1 and 5 use all causes of death. Columns 2 and 6 present estimates that isolate the impact

<sup>25</sup> Event studies using the BJS method differ from OLS event studies in that OLS event studies estimate year-specific coefficients relative to an omitted year. In contrast, BJS event studies do not require an omitted year and instead estimate treatment effects for each lag and lead separately by comparing treated observations  $x$  years before or after the introduction to control (not-yet-treated) observations.

<sup>26</sup> Our primary specification includes these vaccines since the inactivated polio vaccine and the pentavalent vaccine are supported by Gavi and cover the same diseases (respectively) and survey data may combine coverage rates for these two vaccines in a catchall polio category or hepatitis B category.

<sup>27</sup> Between 2001 and 2019, the Measles and Rubella Initiative spent more than US\$1.2 billion supporting measles vaccination in 88 countries. Since we find similar results excluding the measles and rubella coverage results, we conclude that the initiative's activities are not driving our results. It is also worth noting that Gavi is one of the initiative's key supporters: <https://measlesrubellainitiative.org/learn/about-us/>.

TABLE 6—IMPACT OF GAVI-FUNDED INTRODUCTION OF A VACCINE ON CHILD MORTALITY FROM PRIMARILY LINKED CAUSES

| Estimation method                                                                       | OLS                                              |                 |                                        |                                         | BJS                                              |                 |                                        |                                         |
|-----------------------------------------------------------------------------------------|--------------------------------------------------|-----------------|----------------------------------------|-----------------------------------------|--------------------------------------------------|-----------------|----------------------------------------|-----------------------------------------|
|                                                                                         | Causes of death related to a Gavi-funded vaccine |                 |                                        |                                         | Causes of death related to a Gavi-funded vaccine |                 |                                        |                                         |
|                                                                                         | All causes of death (1)                          | All four (2)    | Respiratory and diarrheal diseases (3) | Measles and meningitis/encephalitis (4) | All causes of death (5)                          | All four (6)    | Respiratory and diarrheal diseases (7) | Measles and meningitis/encephalitis (8) |
| <i>Panel A. All countries</i>                                                           |                                                  |                 |                                        |                                         |                                                  |                 |                                        |                                         |
| Postintroduction                                                                        | −1.15<br>(0.57)                                  | −1.19<br>(0.57) | −1.28<br>(0.63)                        | 0.02<br>(0.08)                          | −1.08<br>(0.39)                                  | −1.08<br>(0.39) | −1.14<br>(0.16)                        | −0.05<br>(0.01)                         |
| Observations                                                                            | 28,028                                           | 8,624           | 4,312                                  | 4,312                                   | 27,908                                           | 8,495           | 4,126                                  | 4,303                                   |
| R <sup>2</sup>                                                                          | 0.85                                             | 0.89            | 0.95                                   | 0.75                                    |                                                  |                 |                                        |                                         |
| Dep. var. mean                                                                          | 0.40                                             | 0.32            | 0.57                                   | 0.08                                    | 0.40                                             | 0.30            | 0.53                                   | 0.08                                    |
| <i>Panel B. Gavi recipient or 2000 GNI/GDP per capita &lt; 10,000 (2010 US dollars)</i> |                                                  |                 |                                        |                                         |                                                  |                 |                                        |                                         |
| Postintroduction                                                                        | −1.05<br>(0.56)                                  | −1.08<br>(0.57) | −1.26<br>(0.63)                        | 0.02<br>(0.08)                          | −0.97<br>(0.39)                                  | −0.96<br>(0.38) | −1.01<br>(0.16)                        | −0.06<br>(0.01)                         |
| Observations                                                                            | 15,210                                           | 4,680           | 2,340                                  | 2,340                                   | 15,090                                           | 4,551           | 2,154                                  | 2,331                                   |
| R <sup>2</sup>                                                                          | 0.85                                             | 0.89            | 0.94                                   | 0.75                                    |                                                  |                 |                                        |                                         |
| Dep. var. mean                                                                          | 0.54                                             | 0.50            | 0.89                                   | 0.11                                    | 0.53                                             | 0.46            | 0.84                                   | 0.11                                    |
| <i>Panel C. Gavi recipient or 2000 GNI/GDP per capita &lt; 3,500 (2010 US dollars)</i>  |                                                  |                 |                                        |                                         |                                                  |                 |                                        |                                         |
| Postintroduction                                                                        | −0.96<br>(0.53)                                  | −1.00<br>(0.54) | −1.27<br>(0.62)                        | 0.01<br>(0.10)                          | −0.92<br>(0.37)                                  | −0.89<br>(0.36) | −0.91<br>(0.16)                        | −0.07<br>(0.02)                         |
| Observations                                                                            | 7,007                                            | 2,156           | 1,078                                  | 1,078                                   | 6,887                                            | 2,027           | 892                                    | 1,069                                   |
| R <sup>2</sup>                                                                          | 0.87                                             | 0.89            | 0.94                                   | 0.73                                    |                                                  |                 |                                        |                                         |
| Dep. var. mean                                                                          | 0.61                                             | 0.68            | 1.21                                   | 0.14                                    | 0.59                                             | 0.61            | 1.15                                   | 0.14                                    |

*Notes:* This table presents estimates of the impact of the Gavi-funded introduction of a vaccine on child mortality (deaths of children aged 1–59 months per 1,000 live births) from primarily linked causes. Observations are at the *country × cause of death × year* level and restricted to countries with VR data in the GHO data. Panel A includes data on 122 countries, panel B includes only the 72 countries with GNI/GDP per capita in 2000 < 10,000 (2010 US dollars), and panel C includes the 39 countries with GNI/GDP per capita in 2000 < 3,500 (2010 US dollars). Columns 1–4 provide OLS estimates while columns 5–8 provide estimates using the method in BJS. All regressions include *country × cause of death*, *country × year*, and *cause of death × year* fixed effects. Robust standard errors, clustered by country, are shown in parentheses.

on deaths from vaccine-preventable diseases related to a Gavi-funded vaccine: acute lower respiratory infections, diarrheal diseases, measles, and meningitis/encephalitis. Columns 2 and 6 offer our preferred specification, as other vaccine-preventable causes of death are potentially better counterfactual outcomes than causes of death unrelated to any vaccines. Columns 3 and 7 focus on acute lower respiratory infections and diarrheal diseases, the causes of death targeted by the newer Gavi-funded vaccines (pneumococcal and rotavirus, respectively). Recall from Table 3 that these are the two leading causes of death for children aged 1–59 months. For completeness, columns 4 and 8 provide estimates for the remaining two causes of death related to a Gavi-funded vaccine, measles and meningitis/encephalitis.

Estimates from Table 6 indicate that launches of Gavi-funded vaccines reduce mortality from related causes by about child per 1,000 live births. The results change only slightly when using BJS's estimator (columns 5–8) or limiting the sample to

lower-income countries (panels B and C). Columns 3, 4, 7, and 8 reveal that these effects are driven almost entirely by reductions in respiratory and diarrhea-related deaths. Separate estimates for each affected cause of death support this conclusion (Supplemental Appendix Table A7). Spillovers to unvaccinated children may explain the larger effects on respiratory and diarrheal diseases relative to diseases affected by preexisting vaccines such as measles. The preexisting vaccines had significantly higher coverage rates prior to Gavi introduction (as seen in Figure 3) and were therefore likely closer to herd immunity, when increasing coverage rates would have a smaller effect on mortality.

We note that this substantial reduction in child mortality from respiratory and diarrhea-related deaths is in line with the 10–20 percentage point increase in coverage that we estimate for the pneumococcal and rotavirus vaccines. Using the BJS estimates (columns 7) and panel B of Table 5 on coverage and Table 6 on mortality, we compare the 18.53 pp effect on coverage of the new vaccines and the reduction of 1.01 deaths per 1,000 live births from respiratory and diarrheal diseases. In other words, 185 additional children out of 1,000 live births were vaccinated when Gavi supported the pneumococcal or rotavirus vaccines, and  $1/185 = 5.4/1,000$  newly vaccinated children would have to have died in the absence of the vaccine to explain a mortality reduction of 1 death per 1,000 live births. (If the vaccine were 50 percent effective, then 10.8 in 1,000 children would have been expected to die.) These numbers are similar to the mean rates of deaths from lower respiratory infections and diarrheal diseases in 2000 across all countries (Table 3, column 1) and below the mean mortality rates for Gavi countries (column 3).<sup>28</sup> A one-child reduction in mortality is smaller than the reduction in mortality from these causes from 2000 to 2019 for Gavi recipients. Finally, this mortality effect is also consistent with Tasslimi et al. (2011), an epidemiological model of the Gavi PCV vaccine program that Kremer, Levin, and Snyder (2020) use to estimate the cost-effectiveness of the pneumococcal vaccine advanced market commitment.<sup>29</sup>

*Robustness of the Impact on Mortality.*—Recall that to avoid imputed data, Table 6 restricts the sample of countries to those whose mortality-by-cause data are based solely on vital registration systems. Since this is a selected sample of countries, all of which are lower- or upper-middle income, it is worth repeating our analysis with

<sup>28</sup> A careful reader may notice a few discrepancies in this comparison. As seen in Table 4, the Hib vaccine can also prevent deaths from lower respiratory deaths, so one might argue that the correct comparison would be column 8 in Table 5, which includes Hib and implies that 7.5/1,000 children would have to have died in the absence of the vaccine (1/133). However, the survey data include very few observations of Hib coverage; thus, we prefer the estimates from column 7. In addition, Tables 5 and 6 rely on different samples of countries since we restrict the GHO data to those with VR data. Supplemental Appendix Table A8 replicates Table 5 using only the coverage data for countries whose mortality data are in the VR subset. Here we estimate a 41 percentage point increase in coverage for the new vaccines, suggesting that 2.4 additional children would have to have died in the absence of the vaccine (1/410). These numbers are similar to the mean mortality rates for these diseases in Gavi countries with VR data in 2000 (Table 3, column 5).

<sup>29</sup> Tasslimi et al.'s (2011, Sec 3.2, 265) model predicts that an immunization program of 1.8 billion doses—600 million children vaccinated with a three-dose course—in Gavi countries would lead to 4.2 to 4.8 million fewer deaths between 2010 and 2019 (Table 4 of their paper). Therefore, they predict 7.5 fewer deaths per 1,000 children vaccinated. This is similar to our finding of 1 fewer death per 185 children vaccinated, or 5.4 fewer deaths per 1,000 children vaccinated. Tasslimi et al.'s (2011) model uses a PCV vaccine efficacy rate of 85 percent, which they calibrate based on a meta-analysis of the literature.

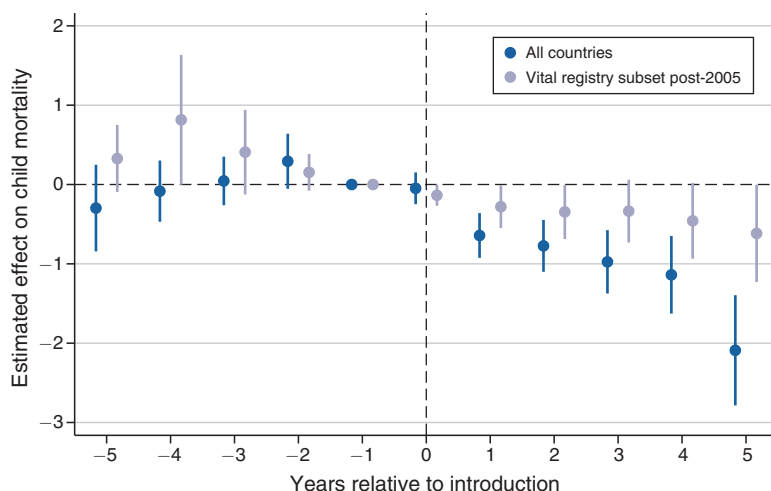


FIGURE 5. IMPACT OF GAVI-FUNDED INTRODUCTION OF A VACCINE ON CHILD MORTALITY BY CAUSE

*Notes:* This figure plots the impact of the Gavi-funded introduction of a vaccine on child mortality rates for primarily linked causes of death. Observations are at the *country*  $\times$  *cause of death*  $\times$  *year* level. Dark navy estimates are using all countries, while light gray estimates restrict the sample to countries with VR data after 2005 (when we have at least ten Gavi recipients with VR data). Only causes of death related to a Gavi-funded vaccine are included in the sample. The regression includes *country*  $\times$  *cause of death*, *country*  $\times$  *year*, and *cause of death*  $\times$  *year* fixed effects. The year before Gavi introduces a given vaccine is the omitted year. Ninety-five percent confidence intervals are estimated from robust standard errors, clustered by country.

all the countries in the GHO data as well. One might expect smaller estimates if the imputations cause classical measurement error, but the estimates could also be larger if Gavi funding and vaccination has larger effects in poorer countries.<sup>30</sup> We find very similar results, qualitatively and quantitatively (Supplemental Appendix Table A9).

Figure 5 presents event-study estimates from specification (4) using both the VR subset of the GHO data (gray) and the full GHO data (dark blue). One caveat to estimating the event study with the VR subset is that the sample of Gavi countries with VR data changes substantially over time (only five Gavi recipients have VR data for the entire 2000–2019 time period). In Figure 5, we restrict the sample to years after 2005 when we have at least ten Gavi recipients with VR data. Following column 2 of Table 6, we include data from only the four causes of death related to a Gavi-funded vaccine. The figure confirms that Gavi funding of a vaccine reduces mortality rates from related causes. The coefficients for the postintroduction years are negative and jointly significant ( $p$ -value  $< 0.01$  for the full sample,  $p$ -value  $< 0.10$  for the VR subset).<sup>31</sup> It is not surprising that the impact grows over time, as it is not until five

<sup>30</sup> The statistical models used by Perin et al. (2022b) use country- and year-specific information such as vaccine coverage rates to estimate mortality by cause, but they do not incorporate information on Gavi support for specific vaccines. While this reduces the risk of circularity in the results, it is possible that the imputations in the full sample inevitably lead us to estimate mortality reductions given coverage increases.

<sup>31</sup> Recall that Figure 4 suggested a two-year lag between a vaccine launch and an impact on cohort-level coverage. Figure 5, on the other hand, suggests only a one-year lag for the impact on child mortality. While this could be explained by catch-up campaigns that target older children, improving mortality rates immediately but also biasing the impact on coverage toward zero, or programs to improve vaccine infrastructure before vaccines are rolled out,

years after introduction that all children under 5 will have been exposed to Gavi's program. Supportive of our identification strategy, we see small, insignificant conditional differences in mortality in the years prior to Gavi funding ( $p$ -value = 0.12 for the full sample,  $p$ -value = 0.4 for the VR subset). BJS's event-study estimator produces a very similar figure (Supplemental Appendix Figure A7).<sup>32</sup>

Supplemental Appendix Figure A8 demonstrates the robustness of our mortality results to a number of variations on the data and specification (results are presented in Supplemental Appendix Tables A10, A11, and A12), including (i) restricting the sample to Gavi recipients; (ii) using vaccine-specific line items from funding data to determine the first year of Gavi funding; (iii) including linear *country*  $\times$  *cause of death* trends or *region*  $\times$  *vaccine*  $\times$  *year* fixed effects; (iv) using the natural log of child mortality (dropping the few zeros); (v) estimating the impact on total under-5 mortality, adding in deaths among infants 0–27 days old; (vi) restricting the sample to countries with VR data for the full time period (2000–2019); and (vii) using GBD mortality data, which uses different modeling procedures than the full sample GHO data.

We also confirm that our results are robust to collapsing the cause-specific mortality data into deaths targeted by Gavi-funded vaccines (acute lower respiratory infections, diarrheal diseases, measles, and meningitis/encephalitis) and other deaths (injuries, malaria, etc.) and then estimating the effect of any Gavi vaccine support as the change in child mortality from vaccine-affected causes relative to nonaffected causes (Supplemental Appendix Table A13). We also look at the impact of Gavi support for any vaccine on all-cause under-5 mortality from the World Development Indicators. We find large decreases in under-5 mortality, but note that here we have to rely on country trends (instead of *country*  $\times$  *year* fixed effects) to deal with time-varying differences across countries (also in Supplemental Appendix Table A13). Finally, estimates of specification (5), allowing each vaccine to affect any cause of death that could reasonably be connected to a given disease, as discussed in Section IID, are qualitatively consistent with our main results (Supplemental Appendix Table A14).

### C. 2SLS Estimates of Gavi's Impact

As discussed in Section IIB, country- and vaccine-specific initiatives correlated with Gavi support have the potential to bias our results. One way to address this possibility is to focus on the variation in the Gavi-funded introduction of a vaccine that is externally driven, based on global vaccine supply constraints or changes in Gavi's total funding or global priorities. Table 7 presents estimates of the impact of Gavi funding on coverage (column 1) and child mortality from causes of death related to a Gavi-funded vaccine for all countries (column 2) and the VR subset (column 3) using country eligibility (based on Gavi's criteria determined in 1999)

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it may also be due to heterogeneous effects of vaccines on child mortality. Specifically, small initial coverage increases may have a substantial effect on mortality if those specific vaccines have a large effect on mortality. To investigate this possibility, Supplemental Appendix Figure A6 reestimates Figure 4 using only the vaccines related to causes of death in the GHO mortality data (rotavirus, measles, pneumococcal, and Hib from the pentavalent vaccine—there are no coverage data for meningitis or encephalitis). Here we see a more pronounced effect on coverage rates in year 1.

<sup>32</sup> We are unable to estimate a BJS event study with the VR subset due to insufficient data.



TABLE 7—2SLS ESTIMATES OF THE IMPACT OF GAVI-FUNDED INTRODUCTION OF A VACCINE

| Outcome                                                                                 | Child mortality from causes of death related to a Gavi-funded vaccine |               |               |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------|---------------|
|                                                                                         | Coverage                                                              |               |               |
|                                                                                         | All vaccines                                                          | All countries | VR subset     |
| Sample                                                                                  | (1)                                                                   | (2)           | (3)           |
| <i>Panel A. All countries</i>                                                           |                                                                       |               |               |
| Postintroduction                                                                        | 8.5<br>(2.6)                                                          | −5.6<br>(1.0) | −2.8<br>(1.7) |
| Observations                                                                            | 6,575                                                                 | 15,112        | 8,600         |
| Cragg-Donald Wald <i>F</i> -statistic                                                   | 885.7                                                                 | 1,482.5       | 777.4         |
| Dep. var. mean                                                                          | 80.3                                                                  | 2.4           | 0.3           |
| <i>Panel B. Gavi recipient or 2000 GNI/GDP per capita &lt; 10,000 (2010 US dollars)</i> |                                                                       |               |               |
| Postintroduction                                                                        | 6.2<br>(2.9)                                                          | −5.3<br>(1.1) | −2.5<br>(1.7) |
| Observations                                                                            | 6,021                                                                 | 11,088        | 4,656         |
| Cragg-Donald Wald <i>F</i> -statistic                                                   | 714.9                                                                 | 789.0         | 367.6         |
| Dep. var. mean                                                                          | 79.4                                                                  | 3.2           | 0.5           |
| <i>Panel C. Gavi recipient or 2000 GNI/GDP per capita &lt; 3,500 (2010 US dollars)</i>  |                                                                       |               |               |
| Postintroduction                                                                        | 5.9<br>(2.9)                                                          | −5.5<br>(1.5) | −2.9<br>(2.1) |
| Observations                                                                            | 4,933                                                                 | 7,888         | 2,132         |
| Cragg-Donald Wald <i>F</i> -statistic                                                   | 445.3                                                                 | 248.6         | 97.7          |
| Dep. var. mean                                                                          | 79.4                                                                  | 4.1           | 0.7           |

*Notes:* This table presents two-stage least squares (2SLS) estimates of the impact of the Gavi-funded introduction of a vaccine using country eligibility interacted with global Gavi funding for a given vaccine in a given year (minus own country receipts) as the instrument. The sample and control variables in column 1 are the same as Table 5, column 1. The sample and control variables in columns 2 and 3 are the same as Table 6, column 2, except for the sample of countries as indicated in the header. Robust standard errors, clustered by country, are shown in parentheses.

interacted with global Gavi funding for a given vaccine in a given year (minus own country receipts) as an instrument for receiving Gavi funding. Thus, this variation is exogenous to recipient-country initiatives, regardless of whether they are vaccine-specific or not. The instrument is not strong enough to separate new and preexisting vaccines or related causes of death, but it offers enough predictive power to estimate our overall results. We find qualitatively similar results: point estimates that are much larger in magnitude than our main results and statistically significant for coverage and mortality using the full GHO sample, but imprecise when we focus on the VR subset and restrict the set of countries by income.

#### *D. Spillover Impacts of Gavi Funding on Nonfunded Vaccines and Unrelated Causes of Death*

In this subsection, we explore the possibility of spillovers from Gavi support for a particular vaccine to nonfunded vaccines, those we use as control vaccines. Like many analyses of foreign aid or government policies, our approach of comparing across vaccines is vulnerable to bias from substitution across outcomes. In this context, the spillovers could be either positive or negative, and from the demand or

TABLE 8—IMPACT OF GAVI-FUNDED INTRODUCTION OF ANY VACCINE ON COVERAGE OF OTHER VACCINES AND UNRELATED CAUSES OF DEATH

|                                                                                         | Coverage                           |                | Causes of death           |                       |
|-----------------------------------------------------------------------------------------|------------------------------------|----------------|---------------------------|-----------------------|
|                                                                                         | All vaccines not yet funded<br>(1) | BCG<br>(2)     | Not (yet) affected<br>(3) | Never affected<br>(4) |
| <i>Panel A. All countries</i>                                                           |                                    |                |                           |                       |
| Post any introduction                                                                   | 0.7<br>(1.5)                       | 0.7<br>(1.4)   | 0.008<br>(0.01)           | −0.007<br>(0.01)      |
| Observations                                                                            | 7,270                              | 1,138          | 27,769                    | 19,404                |
| $R^2$                                                                                   | 0.73                               | 0.84           | 0.85                      | 0.84                  |
| Dep. var. mean                                                                          | 76.3                               | 83.5           | 0.39                      | 0.44                  |
| <i>Panel B. Gavi recipient or 2000 GNI/GDP per capita &lt; 10,000 (2010 US dollars)</i> |                                    |                |                           |                       |
| Post any introduction                                                                   | 0.8<br>(1.5)                       | 0.6<br>(1.4)   | −0.01<br>(0.01)           | −0.01<br>(0.02)       |
| Observations                                                                            | 6,713                              | 1,096          | 14,951                    | 10,530                |
| $R^2$                                                                                   | 0.72                               | 0.82           | 0.86                      | 0.85                  |
| Dep. var. mean                                                                          | 75.2                               | 83.6           | 0.53                      | 0.56                  |
| <i>Panel C. Gavi recipient or 2000 GNI/GDP per capita &lt; 3,500 (2010 US dollars)</i>  |                                    |                |                           |                       |
| Post any introduction                                                                   | 0.7<br>(1.5)                       | −0.05<br>(1.4) | 0.0005<br>(0.02)          | −0.02<br>(0.02)       |
| Observations                                                                            | 5,624                              | 954            | 6,748                     | 4,851                 |
| $R^2$                                                                                   | 0.73                               | 0.83           | 0.88                      | 0.87                  |
| Dep. var. mean                                                                          | 74.4                               | 82.9           | 0.58                      | 0.58                  |

*Notes:* This table presents OLS estimates of the impact of the Gavi-funded introduction of any vaccine on coverage rates for vaccines not funded by Gavi yet using the WHO/UNICEF-collected survey data (columns 1–2) and on child mortality (deaths of children aged 1–59 months per 1,000 live births) from unrelated causes (columns 3–4). The data used in columns 1–2 are the same as in Table 5; the data used in columns 3–4 are the same as in Table 6. Regressions include *country*  $\times$  *vaccine* and *vaccine*  $\times$  *cohort* fixed effects (columns 1–2) or *country*  $\times$  *cause of death* and *cause of death*  $\times$  *cohort* fixed effects (columns 3–4). All regressions include country-specific trends. Robust standard errors, clustered by country, are shown in parentheses.

supply side. For example, when inoculating children using a Gavi-funded vaccine, parents or doctors may be motivated to vaccinate children against other diseases. Alternatively, since Gavi requires cost-sharing, governments may choose to substitute money away from vaccines that they were already supporting toward vaccines that Gavi is helping introduce. If this were the case, the increase in vaccine coverage after introduction could be driven as much by reductions in coverage of other vaccines as by increases in coverage of Gavi-funded vaccines.

We address this possibility by estimating the impact of the introduction of a country's first Gavi-funded vaccine on coverage of vaccines that have not (yet) been introduced by Gavi in that particular country. If our previous results were driven by substitution across vaccines, we would expect to see a negative impact of Gavi funding for any vaccine on coverage for other vaccines. We see no support for this concern. Column 1 of Table 8 estimates the change in vaccine coverage when Gavi first introduces any vaccine (not including the introduced vaccine itself), and column 2 focuses on BCG vaccine coverage since this vaccine, for tuberculosis, has never been funded by Gavi. None of the estimates are statistically significant—in fact, five

of the six point estimates are positive—suggesting that increases in the coverage of Gavi-funded vaccines did not come at the expense of other non-Gavi-funded vaccines; however, given the large standard errors, we cannot rule out the possibility of negative spillover effects.<sup>33</sup>

Supplemental Appendix Figure A9 plots event-study estimates of the impact of receiving any Gavi funding on coverage for vaccines that Gavi has not yet started funding, while Supplemental Appendix Figure A10 plots an event study for the impact of receiving any Gavi funding on coverage of the BCG vaccine. Year 0 is the first year any vaccine was introduced by Gavi in this country. Both figures confirm the results from Table 8 that, if anything, vaccine coverage of non-Gavi-funded vaccines was increasing after the introduction of Gavi-funded vaccines. Given that Gavi funding for a particular vaccine was often paired with funding for other health care and vaccine delivery infrastructure, it is not surprising that coverage for these unfunded vaccines may have risen. In addition, any demand-side interventions to increase coverage may have led to children receiving multiple vaccines, including those not funded by Gavi. These marginal effects could also indicate that Gavi funding freed up funding for countries to support *non-Gavi-funded* vaccines, but the results are not robust enough to make any firm conclusions.

We emphasize that the positive trend in non-Gavi-funded vaccine coverage after Gavi introduction does not detract from our main result that Gavi funding improved vaccine coverage. This conclusion is again due to the *country*  $\times$  *cohort* fixed effects in our preferred specification, which flexibly control for the overall trends in a country's vaccine coverage. The improvements we estimated in Table 5 are in addition to these general improvements in vaccine coverage, leading them to possibly underestimate Gavi funding, if Gavi funding positively impacts this general trend. The results in Table 8 do suggest, however, that our main result is not simply crowd-out of other vaccines by Gavi-funded vaccines. This result allows the possibility that Gavi funding crowds in additional resources from recipient countries, and we note that our effects should be interpreted as the impact of Gavi spending plus additional country spending in response to Gavi programs.

We also address the possibility of such spillovers with respect to child mortality. As with vaccine coverage, it is possible that reductions in child mortality from vaccine-related causes may come at the expense of child deaths from other causes. We explore this issue in columns 3 and 4 of Table 8, where we estimate the impact of any Gavi support on mortality from causes of death unaffected by currently funded vaccines (column 3) or causes of death that are never affected by any of the Gavi-supported vaccines, such as injuries or malaria (column 4). We see small and statistically insignificant effects on child mortality from unrelated causes when Gavi introduces any vaccine, suggesting that improvements in mortality from Gavi-related causes are not coming at the expense of mortality from unrelated causes. We conclude that we find no evidence of crowd-out of other health funding that would affect child mortality.

<sup>33</sup> Note that we cannot include *country*  $\times$  *cohort* fixed effects in the regressions in Table 8, since doing so absorbs all variation in Gavi funding, but we do include country-specific trends.

#### IV. Cost-Benefit Analysis

While we have demonstrated that Gavi funding increased vaccine coverage and reduced child mortality, the cost of the program, US\$14 billion through 2019, is substantial. An important question to address is whether the benefits of Gavi's programs outweigh the cost.

To answer this question, we first calculate lives saved from the Gavi program in country  $i$  at year  $t$  from vaccine  $v$  as follows:

$$LivesSaved_{ivt} = -\Delta mortality \times \frac{Birth\ Cohort_{it}}{1,000} \times Treated_{ivt}.$$

Here,  $-\Delta mortality$  is our estimated reduction in child mortality per 1,000 live births, from Table 6, caused by Gavi introducing funding for a specific vaccine (1 death/1,000 live births). This estimate is constant across cohorts, countries, and vaccines.  $Birth\ Cohort_{it}$  is the size of country  $i$ 's birth cohort at time  $t$ .  $Treated_{ivt}$  is a binary variable indicating that Gavi has introduced vaccine  $v$  in country  $i$  on or before time  $t$ .

We also consider a value of statistical life (VSL) approach to attach a monetary value to the number of lives saved. While we note that the VSL approach is controversial, we believe that it can be an important guide to policymakers when comparing lifesaving projects to nonlifesaving but valuable projects like increased education or infrastructure projects to increase economic output. Along these lines then, we calculate

$$Value_{ivt} = LivesSaved_{ivt} \times Life\ Expectancy\ at\ 5_{it} \times VSLY_{it},$$

where  $Life\ Expectancy\ at\ 5_{it}$  is the life expectancy of a five-year-old in country  $i$  at time  $t$  and  $VSLY_{it}$  is the value of a statistical life year (VSLY) in country  $i$  and time  $t$ .<sup>34</sup> To estimate  $VSLY_{it}$ , we use per capita income for country  $i$  at time  $t$ .<sup>35</sup> We use per capita income as a conservative estimate and three times per capita income as a high estimate, consistent with the literature. For example, Jamison et al. (2013) review the evidence on VSLY in low-income countries and conclude that a reasonable estimate is two to three times per capita income. Chang et al. (2017) argue that per capita GDP should be a lower bound on VSLY that ranges from one to three times per capita GDP.

To calculate the total benefits, we sum across all time periods, countries, and vaccine observations. We find that Gavi funding led to approximately 1.5 million lives saved (estimate: 1.6 million, 90 percent CI: 561,000–2.75 million),<sup>36</sup> at a cost

<sup>34</sup> We derive this formula assuming that  $VSLY_{it}$  grows at the rate  $(1 + g)$  and that the individual time discount factor is  $1/(1 + r)$ , where  $r = g$ .

<sup>35</sup> Data on life expectancy at age 5 and number of births are obtained from the UN Department of Economic and Social Affairs Population Division and are available at <https://population.un.org/wpp/>. Data on per capita GDP are obtained from the World Bank national accounts data and are available at <https://data.worldbank.org/>. Data are adjusted to 2019 US dollars using the CPI.

<sup>36</sup> We calculate 90 percent confidence intervals from a bootstrap procedure using 5,000 draws of our simulated effect on child mortality from a normal distribution with mean equal to one and a standard deviation of 0.4, consistent with the estimate in Table 6, panel B, column 5. We report confidence intervals (as opposed to standard errors) because the distribution of some reported statistics—e.g., cost per life saved—are nonnormal.

of about \$9,000 per life saved (estimate: \$8,750, 90 percent CI: \$5,035–\$23,353). In terms of cost per life-year saved (obtained by multiplying  $Lives\ Saved_{it}$  by  $Life\ Expectancy\ at\ 5_{it}$ ), we find that Gavi spent about \$150 (estimate: \$167, 90 percent CI: \$100–\$463) to save a life-year. The WHO categorizes interventions as extremely cost-effective if they save a life year in a country at a cost of one year of that country's per capita GDP. We find that Gavi easily meets this threshold, as it spends about 20 percent (estimate: 18 percent, 90 percent CI: 11–50 percent) of the recipient country's per capita income to save a life-year.

Using our smallest VSLY estimate, the value of Gavi's program is around US\$150 billion (estimate: US\$153 billion, 90 percent CI: \$52–\$253 billion); using our largest VSLY estimate, it is approximately US\$450 billion (estimate: \$458 billion, 90 percent CI: \$155–\$758 billion). Given that the overall budget for Gavi was US\$14 billion, this represents a return on investment of somewhere between \$10 and \$30 per dollar spent (estimates: \$11, 90 percent CI: \$3.6–\$18, and \$33, 90 percent CI: \$11–\$54).

We believe the above calculation is the correct return calculation from the perspective of Gavi. To calculate the return from a broader perspective, one may wish to include vaccination delivery costs, specifically those not funded by Gavi. Per-dose vaccine costs are estimated to range from US\$1.87 to US\$2.54 (Portnoy et al. 2022; Sim et al. 2021; Vaughn et al. 2019). Based on these delivery costs, we estimate the cost of delivering Gavi vaccines at US\$12 billion.<sup>37</sup> Given that our lowest estimate of the benefits of the Gavi program is US\$153 billion and that Gavi spent US\$14 billion, adding even the entirety of these delivery costs does not change our conclusion that Gavi was cost-effective. Moreover, some of these delivery costs are likely paid by Gavi, as they have spent US\$1.4 billion on health system strengthening.<sup>38</sup>

Next, we compare Gavi's cost-effectiveness to other health interventions in LMI countries, with the caveat that we do not always have causally identified effects of other health development policies. In this paper, our goal is to estimate a plausibly causal effect that often leads to smaller (but more accurate) estimates of the benefits of the evaluated program. Therefore, the following comparison may understate the relative cost-effectiveness of Gavi.

We calculate that Gavi spends \$167 per life year saved. Horton et al. (2017) rank a large number of health interventions using this metric. According to their ranking, Gavi is similar in cost-effectiveness to household water treatment, oral rehydration therapy, and access to contraception. It is also ten times cheaper than their estimate for increased urban water supply sanitation; however, it is more expensive than neonatal care at home and treating malaria with artesunate.

Hendren and Sprung-Keyser (2020) have made a concerted effort to estimate the "marginal value of public funds" (MVPF, a benefit-to-cost ratio) based on causal

<sup>37</sup> We estimate the number of vaccinations that Gavi initiates for a given country  $i$ , vaccine  $v$ , and time  $t$  as  $Vaccinations_{ivt} = Birth\ Cohort_{it} \times Treated_{ivt}$ . Here,  $Treated_{ivt}$  is a binary variable indicating that Gavi has introduced vaccine  $v$  in country  $i$  on or before time  $t$ . We then sum across all vaccines, countries, and time periods to get the total number of Gavi-initiated vaccines, 1.6 billion. Consistent with WHO guidelines (WHO 2023), we assume three doses per vaccination. Multiplying by a per-dose cost of US\$2.5 leads to total delivery costs of US\$12 billion. This estimate is consistent with Sim et al. (2021), who estimate the cost of vaccine delivery for a total of 94 LMI countries between 2011 and 2020 at US\$13.6.

<sup>38</sup> <https://www.gavi.org/sites/default/files/All-Countries-Commitments-and-Disbursements-26112019.xlsx>.

estimates of various programs. However, their estimates are only for US programs. Nevertheless, we estimate the MVPF of Gavi at 11. This number is similar to the introduction of Medicaid, better than Head Start, and about half as big as the District of Columbia Tuition Assistance Grant Program.

## V. Discussion and Conclusion

The role that international organizations can have in improving well-being in LMI countries has been much debated in economics. In this paper, we show that Gavi—an alliance aimed at expanding access to vaccines in low-income countries—has had a substantial positive effect on countries that receive its support.

By exploiting the differential timing in Gavi support across countries and vaccines, we estimate that Gavi's programs increased vaccine coverage rates by 2–5 percentage points on average, with larger effects of 10–20 percentage points for the newer vaccines—pneumococcal and rotavirus—that were generally unavailable in developing countries prior to Gavi's intervention. Additionally, we find that Gavi support has led to an average decrease in child mortality from related causes of 1 child per 1,000 live births. Importantly, we find a reduction in respiratory deaths that could be related to pneumococcal pneumonia and diarrheal deaths that could be related to rotavirus infection. We estimate the narrow economic returns of these mortality gains to be at least \$11 per \$1 spent by Gavi. We view this as a lower bound on the return, as our analysis does not estimate benefits such as vaccine-induced reduced morbidity, Gavi's health system strengthening support, the intrinsic value of human life not captured by the economic VSL measure, and potential feedback from increased health to productivity (see, for example, Bloom, Canning, and Weston 2005 and Summan, Nandi, and Bloom 2022).

Our work has left some important questions unanswered. For example, we are unable to determine whether Gavi's funding model is sustainable or whether donors and countries will be called upon to increase their contributions in the future. Moreover, we cannot determine what the correct proportion of funding should be for health system strengthening versus vaccine purchase, an important policy question for Gavi. It is also clear that Gavi is replicating some actions countries would have already taken (for example, support for vaccines that already have relatively high coverage rates, such as DTP), so it may be possible for Gavi to more efficiently use its resources. Finally, the Gavi approach is a top-down approach. It is unclear the extent to which support for local-based solutions could better address health disparities.

The success of Gavi in channeling international donor funds into beneficial outcomes for recipient countries raises many questions regarding the potential for foreign aid. Specifically, which attributes of Gavi are essential to its success? The focus on vaccination seems key, but there are likely other efficacious large-scale interventions that can be supported with foreign aid. The responsibility assigned to the recipient country, particularly in applying for and cofinancing the interventions, may be important as well. Finally, the unique public-private alliance structure of Gavi may explain, in part, why it succeeded where other global vaccination campaigns did not. We leave these compelling questions to future research.



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